

AKADEMIA GÓRNICZO-HUTNICZA IM. STANISŁAWA STASZICA W KRAKOWIE
WYDZIAŁ FIZYKI I INFORMATYKI STOSOWANEJ

KATEDRA FIZYKI MATERII SKONDENSOWANEJ

Praca dyplomowa

Application of quantum computing to electronic structure calculation using Green's function method

Wykorzystanie komputerów kwantowych do obliczeń struktury elektronowej przy użyciu metody funkcji Greena.

Autor: Martyna Migdałek
Kierunek studiów: Fizyka Techniczna
Opiekun pracy: dr hab. inż. Jakub Haberko

Kraków, 2026


AI

I would like to acknowledge the exceptional guidance provided by PhD Tomasz Stopa from IBM Software Lab, Krakow, Poland whose support was instrumental in the completion of this work. I would also like to express my sincere gratitude to DSc Jakub Haberko for his invaluable assistance and insightful scientific feedback throughout the course of this research.

Abstract

This research applies quantum computing to solve problems related to Multiple Scattering Theory (MST). A one-dimensional implementation is selected due to its key properties and direct analogy to higher-dimensional cases, serving as a foundation for future extensions. Methods incorporating Variational Quantum Deflation (VQD) alongside a custom algorithm are designed and applied to thoroughly explore the utility of secular matrix calculations using quantum algorithms. Significant effort is devoted to the integration of the proposed problem formulation with the Quantum Information Software Kit (Qiskit) programming library to use IBM quantum accessible backends to find the energy of the system. Additionally, visual validation of the results is utilized in the form of wavefunction plotting. All created tools are gathered in a publicly accessible library. The computational complexity of this approach is evaluated in the context of current technological limitations and their planned evolution. Directions for future development of the project are also outlined.

Contents

Abstract **4**

List of Acronyms **7**

I Theory **9**

1 Introduction **10**

2 Multiple scattering method **12**
2.1 Introduction to MST 12
2.2 Mathematical framework of MST 13
2.2.1 Transition from 3D to 1D in MST 13
2.2.2 Green's function in MST 13
2.2.3 An arbitrary set of square wells 15

3 Perspectives on complexity **19**

4 Quantum computing **22**
4.1 Introduction to quantum information 22
4.2 Overview of quantum diagonalization algorithms 25
4.2.1 Variational Quantum Eigensolver and Variational Quantum Deflation 25
4.2.2 Other methods of quantum diagonalization 26
4.3 The Qiskit framework 26
4.3.1 Backends 26
4.3.2 Software tools and Qiskit modules 28

5 Quantum algorithmic approach to electronic structure theory **30**
5.1 Form of the secular matrix 30
5.2 Decomposition of the secular matrix into Pauli terms 31
5.3 Implementing VQD 31

II Results **32**

6 Computational results overview **33**
6.1 Reference classical calculations 33

6.2 Comparison of VQD results 37
6.3 Results verification - plotting wavefunctions 42

7 Novel quantum algorithm for MST problem solution 45

8 Scalability and complexity gain analysis 48
8.1 Time and space resource consumption in NumPy diagonalization . . 48
8.2 Quantum computation scaling 49

9 Architecture and Tools of the QuantumMST Framework 51

10 Future outlook 54

Summary 57

Bibliography 57

Appendix A Potential Distribution of the 21-Scatterer System 62

List of Acronyms

BQP bounded-error quantum polynomial time. 21

CLOPS circuit layer operations per second. 27

COBYLA Constrained Optimization by Linear Approximations. 28, 38

EPLG errors per layered gate. 27

FNP Functional Nondeterministic Polynomial time. 20

FP Functional Polynomial time. 20

FTQC Fault-Tolerant Quantum Computing. 11

HPC High-Performance Computing. 11

KKR Korringa-Kohn-Rostoker. 55

KQD Krylov Quantum Diagonalization. 26

MST Multiple Scattering Theory. 4–6, 11–13, 16, 19, 37, 45, 49, 54, 55, 57

MT muffin-tin. 12, 13, 15, 16

NISQ Noisy Intermediate-Scale Quantum. 11, 26

NP Nondeterministic Polynomial time. 19–21, 28

P Polynomial time. 19–21

QCSC Quantum-Centric Supercomputing. 11

QDAs Quantum Diagonalization Algorithms. 25

Qiskit Quantum Information Software Kit. 4, 5, 24, 26, 28, 29, 31, 37, 38, 51, 55, 63

QMA quantum Merlin-Arthur. 21

QPU Quantum Processing Unit. 11, 25, 26, 63

SQD Sample-based quantum diagonalization. 26, 49

VQD Variational Quantum Deflation. 4, 5, 25, 28, 30, 31, 37–42, 45, 46, 49, 51, 57, 63, 64

VQE Variational Quantum Eigensolver. 25, 26, 46, 63

Part I

Theory

1. Introduction

It has been 10 years since IBM quantum computers were placed in the cloud for public access. Since then, 60 quantum devices below 100 qubits and 30+ quantum backends above 100 qubits that are capable of running circuits with more than 5,000 two-qubit gates have been made available.[1] World-class scientists are constantly working to advance this technology, meticulously executing the objectives outlined in their regularly published roadmaps. According to the latest projections, a quantum computer performing 100 million gates across 200 qubits is expected to arrive by 2029. A natural question that arises is why there is such strong interest in this form of computing when we already have such powerful classically driven supercomputers. What makes it so special is that it opens up a new way of processing information by utilizing the laws of quantum mechanics, a direction that is unavailable for ordinary devices. The exponential complexity of simulating quantum systems on classical hardware inspired Feynman,[2] who is considered to be one of the pioneers of quantum computing. The classical information processing is based on bits that are 0 or 1. The quantum approach uses previously mentioned qubits, which can represent a superposition of both states simultaneously. Instead of storing n bits (n numbers), n qubits allow us to store 2^n complex amplitudes. To better visualize this storage capability, we can express it in terms of classical RAM. A single complex number requires 16 bytes (8 bytes each for the real and imaginary parts), which we then multiply by $\frac{2^n}{1024^3}$, where 1024^3 accounts for unit conversion. According to this logic, a typical laptop with 16 GB of RAM can simulate up to 30 qubits, while supercomputers such as IBM Summit[3] can reach around 48 qubits, and all the classical computers on Earth combined could simulate roughly 60 qubits.

However, unlocking the potential hidden in quantum computers is by no means simple. Contrary to popular belief, we are not talking about an all-powerful machine that can solve every possible problem, but rather specific ones where, to achieve a speedup over traditional calculations, superposition, entanglement, or interference must be cleverly used. In spite of numerous algorithm proposals (for examples, see the Catalog of Quantum Algorithms[4]), there exist only several that promise actual speedup, including Grover's algorithm,[5] which targets unstructured search tasks and provides a quadratic (polynomial) speedup, and Shor's algorithm[6] for number factorization, which achieves an exponential (superpolynomial) one. For this reason, researchers are actively searching for new applications of known algorithms as well as developing new ones, as a clear quantum advantage over classical processing has yet to be demonstrated. Key areas benefiting from quantum computing include simulating natural systems, analyzing structured data and mathematics, and solving search or optimization problems. For instance, let us narrow these domains down to chemical problems,[7] specifically catalysis. A catalyst increases the efficiency of

chemical reactions without being directly consumed in the process. Designing a catalyst requires a deep understanding of electronic structure, which allows for an educated guess prior to synthesizing it in the lab.

The main motivation for this work is to explore one of the most important and universal methods for describing the electronic structure, which is Multiple Scattering Theory (MST). Specifically, this involves evaluating the applicability of this method to software dedicated to quantum computing, as well as analyzing the emerging challenges. An effort is made to simulate a benchmark system to verify the correctness of the results and subsequently identify its limits, defined here as simulating the largest possible system and comparing it with classical approaches. Furthermore, this study aims to identify the technological advancements in quantum computing required to achieve a true quantum advantage in this particular problem.

IBM uses superconducting hardware that is very sensitive to environmental noise and temperature fluctuations. Although the single gate error rate is usually around $\sim 0.1\%$, the probability of a faulty output increases with the number of operations, which means that lower error rates allow deeper circuits without computational failure. Due to these hardware limitations, the current stage of quantum computing is known as the Noisy Intermediate-Scale Quantum (NISQ)[8] era. The Fault-Tolerant Quantum Computing (FTQC) era is anticipated around 2029 with the introduction of IBM Quantum Starling, which aims to implement quantum error correction. This will allow real-time detection and correction of errors during circuit execution at the expense of requiring more physical qubits. For now, we are limited to employing error mitigation, which relies on reducing errors in post-processing after results are obtained. This is why forecasts indicate such a huge increase in the number of reliable operations from five thousands to hundreds of millions in 2029. Although this type of physical quantum processor is very unlikely to be entirely error-free, error correction is believed to allow such a large number of operations to be executed while maintaining a high probability of an accurate result.

Currently, the most promising path to breakthroughs requires a hybrid approach that integrates a Quantum Processing Unit (QPU) with classical processors to produce meaningful computational outcomes. In the most demanding applications, Quantum-Centric Supercomputing (QCSC)[9,10] is utilized, integrating High-Performance Computing (HPC)[11] - infrastructure powered by supercomputers - with quantum hardware. Marked by IBM's landmark paper,[12] the field entered quantum utility, defined as the capability of quantum processors to execute quantum circuits that surpass the attempt to emulate those exact circuits by the classical simulator. Nearly three years later, the first proofs of quantum advantage, in which quantum approaches outperform classical ones, are published.[13–16] To stay up to date with emerging problem candidates for quantum hardware, it is recommended to check the Quantum Advantage Tracker,[17] which gathers, verifies, and benchmarks quantum performance against classical methods. Near-term the biggest breakthroughs are promised to be in computational fields like Hamiltonian simulation, machine learning, differential equations, and optimization. More broadly, potential applications are anticipated across multiple industries including healthcare (disease risk prediction, drug discovery), finance (fraud detection), automotive (material design), energy (battery design), and high tech (seismic imaging, supply chain planning), which holds great promise for technological advancement and quality of life.

2. Multiple scattering method

2.1 Introduction to MST

The subject of the discourse is Multiple Scattering Theory (MST), which serves as a powerful tool in various fields of physics. This theory is believed to have originated with Rayleigh. In one of his pioneering works on this topic,[18] he described the behavior of a plane acoustic wave as it encounters a fixed, rigid spherical object and analyzed the subsequent scattered waves. Within the realm of quantum mechanics, it has progressively developed into a method for solving the Schrödinger equation. It can be found in many other classical, statistical, or quantum differential equations, such as the Helmholtz, Debye-Hückel, and Dirac equations. In this paper, this approach is utilized to explore electronic structure calculations.[19]

What makes this method so unique is that individual scattering solutions can be obtained before constructing the global one. This leads to its exceptional versatility, allowing it to be adapted to diverse physical geometries. The direct determination of the Green's function in MST is essential for modeling perturbations such as external fields or impurities, as well as for going beyond the standard single-electron approximation. Furthermore, due to its minimal basis set requirements, the computational cost remains manageable even when dealing with large-scale systems. A widely adopted simplification in electronic structure calculations is called the muffin-tin (MT) approximation. It divides the chosen system into two regions: one inside the atomic spheres, where the potential is spherically symmetric and centered around each atomic nucleus, and another in the interstitial space, where the potential is treated as a constant. This can be written as:

$$V(\vec{r}) = \begin{cases} v_n(|\vec{r} - \vec{R}_n|) & \text{for } |\vec{r} - \vec{R}_n| \leq r_{\mathrm{MT}} \\ v_0 & \text{for } |\vec{r} - \vec{R}_n| > r_{\mathrm{MT}} \end{cases} \tag{2.1}$$

where $\vec{r}$ is the vector in a three-dimensional space, $\vec{R}_n$ represents the position vector of the center of the n-th scatterer, $|\vec{r} - \vec{R}_n|$ is the distance from the center of the n-th atom to the given point, $v_n(|\vec{r} - \vec{R}_n|)$ is the spherically symmetric potential inside the atomic sphere, r_{MT} denotes the muffin-tin radius defining the boundary of the spherically symmetric region, and v_0 is the constant potential value outside the atomic spheres.

To obtain a solution in the form of a secular equation (which will be mentioned in sec. 2.2.3) for this potential model, individual scattering solutions for scatterers within a given geometry must be combined. To achieve such a single-site solution, it is first necessary to calculate all the required information for a single scatterer, for instance, in the form of a scattering matrix. To successfully link these solutions, in-

formation about the system's geometry is required, which is obtained by calculating the structural constants.

2.2 Mathematical framework of MST

2.2.1 Transition from 3D to 1D in MST

Given that the angular momentum expansions in 3D would extend to infinity, in this study I restricted the formalism to 1D.[20] Notably, the solutions for the 1D and 3D cases are entirely analogous. For most of the equations, adapting the formalism to one dimension simply requires reducing the position vector $\vec{R}$ to a single spatial coordinate and truncating the summations exclusively to $\ell = 0$ and $\ell = 1$. As a further simplification, a constant potential is assumed within these symmetric intervals representing the MT potential, yielding a set of disjoint square wells. Within the non-zero potential regions the equation changes in the following way:

$$V(\vec{r}) = \sum_n v_n(\vec{r} - \vec{R}_n) \quad \xrightarrow{\text{3D} \to \text{1D}} \quad V(x) = \sum_n v_n(x - X_n). \tag{2.2}$$

The focus of the calculations is placed on bound states ($E < 0$), for which the Schrödinger equation is to be solved. It is formulated as follows:

$$[-\nabla^2 + V(x) - E]\psi(x, E) = 0, \tag{2.3}$$

where Rydberg units are adopted ($\hbar = 1$ and $2m = 1$), simplifying the kinetic energy operator to $-\nabla^2$, and $V(x)$ is a MT potential, for which the potential in the interstitial region is chosen as the energy reference and set to zero.

2.2.2 Green's function in MST

Instead of directly finding $\psi(x, E)$, a transition was made to a method based on the Green's function, which encodes information about both the energy spectrum and the spatial structure. In its simplest form, the Green's function $G(x, x')$ for an operator $\hat{A}$ is defined by

$$\hat{A}G(x, x') = \delta(x - x'). \tag{2.4}$$

In other words, acting with $\hat{A}$ on $G(x, x')$ yields the Dirac delta function, meaning that $G(x, x')$ acts as the inverse of $\hat{A}$. In the realm of quantum scattering in one-dimention, the relevant operator is $\hat{A} = E - H$, where the Hamiltonian operator is denoted:

$$H = -\nabla^2 + V(x) = -\frac{\mathrm{d}^2}{\mathrm{d}x^2} + V(x), \tag{2.5}$$

which leads to the following condition for Green's function:

$$(E - H)G(x, x') = \delta(x - x'). \tag{2.6}$$

Without the presence of an external potential, so for a free-particle Green's function G_0, it changes to

$$(E - H_0)G_0(x, x') = \delta(x - x'). \tag{2.7}$$

The complete Green function G is expressed in terms of the free-particle propagator G_0 and the total scattering matrix T (from all potentials v_n):

$$G = G_0 + G_0 T G_0. \tag{2.8}$$

The scattering expansion is obtained by expanding matrix T in terms of scattering matrices for a single potential t^n:

$$T = \sum_n t^n + \sum_n \sum_{m \neq n} t^n G_0 t^m + \sum_n \sum_{m \neq n} \sum_{p \neq m} t^n G_0 t^m G_0 t^p + \dots, \tag{2.9}$$

where the first term is the sum of single scattering on atom n, the second term contains single scattering on atom m, free propagation to site n and a subsequent scattering event at atom n, with other terms following analogously. This infinite series is not trivial to calculate, that is why it is required to introduce scattering path operator τ, which allows us to represent Eq. (2.9) as:

$$T = \sum_{nn'} \tau^{nn'}. \tag{2.10}$$

This new τ satisfies the equation

$$\begin{aligned} \tau^{nn'}_{\ell\ell'} &= t^n_\ell \delta_{\ell\ell'} \delta_{nn'} + \\ &+ \sum_{\ell'} \sum_{m \neq n} t^n_\ell G^{nm}_{\ell\ell'} \tau^{mn'}_{\ell_2 \ell'}, \end{aligned} \tag{2.11}$$

where $G^{nm}_{\ell\ell'}$ denote structure constants, which represent the site and momentum dependent matrix elements (definitions of $G^{nm}_{\ell\ell'}$ and t^n_ℓ are elaborated later in Sec. 2.2.3). The site indices are $n, m \in \{1, \dots, N\}$, where N denotes the total number of scatterers, while $\ell, \ell' \in \{0, 1\}$. This relation in matrix notation can be rewritten as:

$$\tau = t + tG\tau, \tag{2.12}$$

or equivalently:

$$\tau^{-1} = t^{-1} - G, \tag{2.13}$$

where τ, t and G are square matrices of dimension $2N \times 2N$. In particular, the diagonal blocks of t represent single-site scattering matrices t^n and the off-diagonal matrix G accounts for scattering between different sites $n \neq m$. The allowed energies correspond to the poles of the Green's function, in which the operator τ diverges. Consequently, the secular condition can be formulated as:

$$\det\left(t^{-1}(E) - G(E)\right) = 0. \tag{2.14}$$

In the next part, this condition will be derived by matching wavefunctions at the boundaries.

2.2.3 An arbitrary set of square wells

The secular matrix

Returning to the Schrödinger equation (2.3), each zone (the interior of the n-th scatterer I_n and the interstitial region II) has its own wavefunction that must be calculated separately. Their formulas in the respective regions are given by:

$$\psi_{I_n} = \sum_{\ell=0,1} c_\ell^n R_\ell^n(r_n) Y_\ell(\hat{r}_n), \tag{2.15}$$

and

$$\psi_{II} = \sum_n \sum_{\ell=0,1} b_\ell^n h_\ell(z_n) Y_\ell(\hat{r}_n). \tag{2.16}$$

with the dimensionless argument $z_n = \sqrt{E} r_n$, which is the product of the square root of the energy and the radius of the n-th potential well. The summation $\sum_\ell$ runs over all possible angular momentum states in 1D, while the summation $\sum_n$ accounts for all scattering centers in the system. It is worth highlighting that r_n and $\hat{r}_n$ denote different quantities: the distance and direction relative to the center of the n scatterer, respectively. The amplitudes c_ℓ^n and b_ℓ^n act as expansion coefficients, determining the weights of their respective components. They account for the contribution of incident and scattered waves. For each center n, $R_\ell^n(r_n)$ represents the radial solution of the Schrödinger equation:

$$\left[-\frac{d^2}{dr_n^2} + v_n(r_n) - E \right] R_\ell^n(E, r_n) = 0, \tag{2.17}$$

while $Y_\ell(\hat{r}_n)$ denotes the spherical harmonics, which in one dimension are given by:

$$Y_0(\hat{r}) = \frac{1}{\sqrt{2}} \quad \text{and} \quad Y_1(\hat{r}) = \frac{r}{|r|\sqrt{2}}. \tag{2.18}$$

The wavefunction in the interstitial region is expressed as a sum of scattered Hankel waves originating from all scattering centers. These one-dimensional functions are defined by:

$$h_\ell(z_n) = j_\ell(z_n) + i n_\ell(z_n), \tag{2.19}$$

where the 1D equivalents of the Bessel and Neumann functions are:

$$j_0^{\mathrm{1D}}(z) = \cos(z) \quad \text{and} \quad j_1^{\mathrm{1D}}(z) = \sin(z), \tag{2.20}$$

$$n_0^{\mathrm{1D}}(z) = \sin(z) \quad \text{and} \quad n_1^{\mathrm{1D}}(z) = -\cos(z). \tag{2.21}$$

Here, the Hankel functions enter as components of the free-space Green's function.

To ensure the continuity of the wavefunction across the boundaries, we introduce the expansion of the interstitial wavefunction locally referenced to a particular scatterer. This expansion consists of both incident and scattered wave components:

$$\psi_{II_n} = \sum_{\ell=0,1} a_\ell^n j_\ell(z_n) Y_\ell(\hat{r}_n) + \sum_{\ell=0,1} b_\ell^n h_\ell(z_n) Y_\ell(\hat{r}_n). \tag{2.22}$$

Consistency between Eqs. (2.22) and (2.15) is achieved by matching the wavefunctions at the MT radius r_n^0. Meanwhile, the amplitude ratio a_L^n / b_L^n ensures that the

logarithmic derivatives remain continuous at the boundaries, which is formulated in terms of the scattering matrix t_ℓ^n. However, in practical calculations, it is more convenient to employ its inverse form of scattering matrix m_ℓ^n. The relation between the coefficients is given by the following:

$$a_\ell^n = m_\ell^n b_\ell^n, \tag{2.23}$$

where $m_\ell^n(E)$ can be explicitly expressed as:

$$m_\ell^n(E) = -\frac{z_n^0 h_\ell'(z_n^0) - r_n^0 \gamma_\ell^n h_\ell(z_n^0)}{z_n^0 j_\ell'(z_n^0) - r_n^0 \gamma_\ell^n j_\ell(z_n^0)}. \tag{2.24}$$

Here, γ_ℓ^n represents the logarithmic derivative of the internal radial wavefunction evaluated at the MT radius r_n^0:

$$\gamma_\ell^n(E) = \left[\frac{R_\ell^{n\prime}(E, r_n)}{R_\ell^n(E, r_n)}\right]_{r_n = r_n^0}. \tag{2.25}$$

A globally valid and coherent description of the interstitial region (II) cannot be achieved by directly applying Eq. (2.22) centered on n-th scatterer. This limitation arises because, for bound states ($E < 0$), Hankel functions exhibit exponential growth as the distance r_n increases, leading to divergence issues outside the local cell. Enforcing consistency between muticentered expansion 2.16 with local single centered one (2.22) gives:

$$\sum_{\ell=0,1} \left[a_\ell^n j_\ell(z_n) + b_\ell^n h_\ell(z_n)\right] Y_\ell(\hat{r}_n) = \sum_{n'} \sum_{\ell'=0,1} b_{\ell'}^{n'} h_{\ell'}(z_{n'}) Y_{\ell'}(\hat{r}_{n'}). \tag{2.26}$$

By substraciting $n' = n$ from both sides we obtain

$$\sum_{\ell=0,1} a_\ell^n j_\ell(z_n) Y_\ell(\hat{r}_n) = \sum_{n' \neq n} \sum_{\ell'=0,1} b_{\ell'}^{n'} h_{\ell'}(z_{n'}) Y_{\ell'}(\hat{r}_{n'}). \tag{2.27}$$

To finalize the derivation of the secular equation in MST, convergent expansions of the following form are required:

$$h_{\ell'}\left(z_{n'}\right) Y_{\ell'}(\hat{r}_{n'}) = \sum_{\ell=0,1} j_\ell\left(z_n\right) Y_\ell(\hat{r}_n) G_{\ell\ell'}^{nn'}. \tag{2.28}$$

Here, the expansion coefficients $G_{\ell\ell'}^{nn'}$, originating from the free-space Green's function, act as a propagator that translates the wave scattered from center n' into the local representation of center n. This structural constant in 1D is reduced to $\ell, \ell' \in \{0, 1\}$, which results in a 2×2 matrix:

$$G^{nn'}(E, X_{nn'}) = \begin{pmatrix} 1 & i\,\mathrm{sgn}(X_{nn'}) \\ -i\,\mathrm{sgn}(X_{nn'}) & 1 \end{pmatrix} \times \exp\left[i(E)^{1/2} R_{nn'}\right], \tag{2.29}$$

where, besides the distance between centers $R_{nn'}$, there is a difference between positions $X_{nn'}$. Inserting Eq. (2.28) into Eq. 2.27 gives:

$$\sum_{\ell=0,1} a_\ell^n j_\ell(z_n) Y_\ell(\hat{r}_n) = \sum_{n' \neq n} \sum_{\ell'=0,1} b_{\ell'}^{n'} \left[\sum_{\ell=0,1} j_\ell(z_n) Y_\ell(\hat{r}_n) G_{\ell\ell'}^{nn'}\right], \tag{2.30}$$

then we can change the order of the sums

$$\sum_{\ell=0,1} j_\ell(z_n) Y_\ell(\hat{r}_n), a_\ell^n = \sum_{\ell=0,1} j_\ell(z_n) Y_\ell(\hat{r}_n) \left[\sum_{n' \neq n} \sum_{\ell'=0,1} G_{\ell\ell'}^{nn'} b_{\ell'}^{n'} \right] \tag{2.31}$$

and compare coefficients on both sides

$$a_\ell^n = \sum_{n' \neq n} \sum_{\ell'=0,1} G_{\ell\ell'}^{nn'} b_{\ell'}^{n'}. \tag{2.32}$$

Physically, the incoming wave arriving at a specific scattering site n is expressed as a sum of outgoing waves scattered from other sites. We can rewrite it according to Eq. (2.23) as:

$$m_\ell^n b_\ell^n = \sum_{n' \neq n} \sum_{\ell'} b_{\ell'}^{n'} G_{\ell\ell'}^{nn'} \tag{2.33}$$

The existence of a non-trivial solution to this system of homogeneous linear equations yields the energy eigenvalues of Eq. (2.3). This compatibility criterion is widely known as the secular equation, assuming that the trivial solution is neglected.

Rearranging all terms to the left side leads to the matrix formulations:

$$\left(\begin{bmatrix} m^A & 0 & 0 & \dots \\ 0 & m^B & 0 & \dots \\ 0 & 0 & m^C & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} - \begin{bmatrix} 0 & G^{AB} & G^{AC} & \dots \\ G^{BA} & 0 & G^{BC} & \dots \\ G^{CA} & G^{CB} & 0 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \right) \begin{bmatrix} b^A \\ b^B \\ b^C \\ \vdots \end{bmatrix} = 0 \tag{2.34}$$

$$\begin{bmatrix} m^A & -G^{AB} & -G^{AC} & \dots \\ -G^{BA} & m^B & -G^{BC} & \dots \\ -G^{CA} & -G^{CB} & m^C & \dots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \begin{bmatrix} b^A \\ b^B \\ b^C \\ \vdots \end{bmatrix} = 0 \tag{2.35}$$

In this symbolic notation, m^A and G^{AB} are 2×2 matrices for each allowed angular momentum channel, while b^A likewise contains two coefficients corresponding to $\ell = 0$ and $\ell = 1$. Since the components of the vector $\vec{b}$ represent the amplitudes of the scattered waves, they cannot all be identically zero to prevent the system from being physically meaningless. Therefore, to ensure the existence of non-trivial amplitudes, the determinant of the matrix must vanish, leading to the condition:

$$\det \begin{vmatrix} m^A & -G^{AB} & -G^{AC} & \dots \\ -G^{BA} & m^B & -G^{BC} & \dots \\ -G^{CA} & -G^{CB} & m^C & \dots \\ \vdots & \vdots & \vdots & \ddots \end{vmatrix} = 0. \tag{2.36}$$

This relation serves as the foundation for the problems investigated in this study and constitutes the starting point for all discussed methods. Hereafter, this system matrix is referred to as the secular matrix, where calculating its determinant and evaluating its zero-crossings determines the eigenvalues of the system.

Wavefunction equation

After calculating the energies from the secular matrix, an important step to verify them is plotting the wavefunctions. When these energies are calculated correctly, the segments of the wavefunction in each region should smoothly join together. In order to do that, we require the inverses of the scattering matrices and the structure constants from the secular matrix, which will be used to determine the b coefficients Eq. (2.35). The calculated values are to be substituted into Eq. (2.22), which can be rewritten using Eq. (2.23) as:

$$\psi_{II_n} = \sum_{\ell=0,1} b_\ell^n \left[h_\ell(z_n) + m_\ell^n j_\ell(z_n)\right] Y_\ell(\hat{r}_n). \tag{2.37}$$

The functions outside the scatterers are to be calculated from Eq. (2.15), where the missing quantities are the amplitudes c. As previously mentioned, we must ensure the continuity of the functions, so to calculate these coefficients we match equations (2.15) and (2.37) at the boundary of the n-th scatterer r_n^0:

$$\sum_\ell c_\ell^n R_\ell^n(r_n^0) Y_\ell(\hat{r}_n^0) = \sum_\ell b_\ell^n \left[h_\ell\left(\sqrt{E}\, r_n^0\right) + m_\ell^0 j_\ell\left(\sqrt{E}\, r_n^0\right)\right] Y_\ell(\hat{r}_n^0) \tag{2.38}$$

$$c_\ell^n R_\ell^n(r_n^0) = b_\ell^n \left[h_\ell\left(\sqrt{E}\, r_n^0\right) + m_\ell^0 j_\ell\left(\sqrt{E}\, r_n^0\right)\right] \tag{2.39}$$

$$c_\ell^n = \frac{b_\ell^n \left[h_\ell\left(\sqrt{E}\, r_n^0\right) + m_\ell^0 j_\ell\left(\sqrt{E}\, r_n^0\right)\right]}{R_\ell^n(r_n^0)} \tag{2.40}$$

Having determined all the required variables, we can calculate the wavefunctions outside and inside the scatterer, respectively.

3. Perspectives on complexity

The complexity of a given system often correlates with the level of scientific interest it generates. When simulating such systems, simplified models that capture their key features are commonly employed. MST is one of the theoretical models utilized to simulate complex structures. Paradigms of complexity diverge significantly across academic fields. Their core approaches to explaining complex phenomena are fundamentally different: while physicists seek to discover underlying laws that explain a given system as simply as possible, computer scientists analyze the so-called worst-case complexity to ensure that an algorithm remains reliable under any circumstances. According to the Nobel Prize-winning physicist Anderson,[21] a complex system is a structure composed of a set of elements that exhibit new behaviors driven solely by their interactions. This collective behavior is known as emergence, which implies that the system cannot be understood simply by analyzing its individual components in isolation. In such system, the elements can either amplify or cancel each other's effects. This leads to a disproportionate relationship between inputs and outputs, a defining trait known as nonlinearity.

A classic illustration of this process can be found in the molecules of a liquid or a solid, which exhibit macro-level properties entirely absent in the particles themselves when isolated. The nature of how this type of structure is formed can also be seen in snowflakes,[22] which create remarkable fractal patterns instead of a shapeless lump. Proteins provide another ideal example. On their own, their individual components, amino acids, exhibit no high-level biological function, even though each possesses distinct physicochemical properties such as electric charge, hydrophobicity, or chemical reactivity. However, when they form long ribbon-shaped chains, they acquire more enhanced functionalities, for example, transporting oxygen molecules in the blood, breaking down nutrients, or inducing muscle contractions. Fundamentally, as additional components (e.g., particles) enter a system, the number of components becomes a catalyst for complexity. The same applies to connectivity, which scales up driven by a rapidly expanding network of interactions. To illustrate, two molecules generate just a single interaction, 10 molecules yield 45 distinct pairs, and a group of 100 molecules results in nearly 5,000 potential interactions.

Shifting the focus to computational complexity, this property is strictly measurable, usually by the amount of resources, including time or RAM, required to perform given operations. It is denoted using big $\mathcal{O}$ notation (e.g., $\mathcal{O}(n)$, $\mathcal{O}(1)$, $\mathcal{O}(\log n)$ or $\mathcal{O}(n^2)$), which symbolizes the scalability of the algorithm or computational problem as the input grows. There are several famous challenges concerning complexity classes. One of them is the question of whether Polynomial time (P) (solvable in polynomial time) and Nondeterministic Polynomial time (NP) (in which a solution is easy to verify but difficult to find) categories are the same.[23] However, P and NP

are decision problems, which means they can be identified with boolean functions. Some issues do not belong to these classes, because we want to obtain a specific result not just a 'yes' or 'no' answer.

Matrix diagonalization[24] required for solving the secular matrix Eq. 2.36 derived above, falls into this category. This fundamental problem in linear algebra, which involves computing both diagonal matrix D containing the eigenvalues and V matrix containing eigenvectors of a square complex matrix $A \in \mathbb{C}^{n\times n}$

$$A = VDV^{-1}, \tag{3.1}$$

is closely related to finding the roots (λ) of the characteristic polynomial:

$$\det(A - \lambda I) = 0. \tag{3.2}$$

According to the Abel-Ruffini theorem, if the matrix size is greater than or equal to five ($n \geq 5$), the corresponding equation is a polynomial of degree five or higher, making it impossible to obtain an exact algebraic result in radicals. Therefore, numerical approximation techniques must be employed. The two main schemes are variable precision rational arithmetic and finite precision arithmetic. The former is widespread in theoretical fields because it stores numbers as fractions $\frac{a}{b}$, which allows us to maintain exact results without any loss of precision. This approach requires storing $\log_2(|a|) + \log_2(|b|)$ bits, though this length can even double in size during a single arithmetic operation. Consequently, the maximum overall complexity is the product of the total number of operations and the cost of the bit-level arithmetic operation of the maximum bit length. However, when considering the eigenvalue problem, irrational numbers appear, which cannot be expressed as fractions, meaning that working with approximations remains unavoidable. The second scheme, utilized mainly in numerical analysis, immediately introduces an approximation of a specific accuracy. This method comes with a drawback, as every operation may reduce the accuracy of the final result due to approximation. Regarding complexity, the total bit complexity is equal to the number of operations multiplied by $O^*(b)$, where b is fixed required bit precision.

As mentioned above, the complex eigenvalue problem does not fit the standard P or NP classes. Instead, problems like this can be categorized in an alternative way, as Functional Polynomial time (FP) or Functional Nondeterministic Polynomial time (FNP),[25] where a specific output must be calculated rather than a binary decision. In practice, it is common to simply refer to them as P or NP. In finite arithmetic,[24] the computational time complexity of approximately diagonalizing an arbitrary square matrix under the backward error criterion can be expressed as:

$$\mathcal{O}\left(T_{MM}(n)\log^2\left(\frac{n}{\delta}\right)\right), \tag{3.3}$$

where n is the dimension of the matrix, δ represents the target backward error parameter, and $T_{MM}(n)$ denotes the computational cost of numerically stable multiplication of two $n \times n$ matrices, which satisfies the condition:

$$T_{MM}(n) = \mathcal{O}(n^{\omega+\alpha}) \tag{3.4}$$

for any $\alpha > 0$. Here, ω stands for the exponent of matrix multiplication, which ranges from 3 for the standard algorithm down to approximately 2.37 for fast algorithms.

The factor $\log^2\left(\frac{n}{\delta}\right)$ represents the number of matrix multiplications required by the algorithm. The classification of this problem will be evaluated in greater detail in a later discussion, based on the type of algorithm employed by a given numerical solution.

A completely different classification into complexity classes is adopted in the case of quantum algorithms.[26] The quantum counterpart of P is the bounded-error quantum polynomial time (BQP) class. Similarly to the previous cases, this represents the class of the simplest problems solvable in polynomial time, but within the framework of a quantum computer. Analogously for NP there is quantum Merlin-Arthur (QMA) class, which involves problems whose solutions can be verified in polynomial time using a quantum state as a proof. While NP and QMA define classes of verifiable problems, in general, their most difficult instances are formally designated as NP-complete and QMA-complete. Problems that are at least as hard as these, yet do not necessarily belong to NP or QMA, are referred to as NP-hard and QMA-hard.

In summary, the issue of complexity can be approached from multiple perspectives, illustrating the true depth and multifaceted nature of the topic. Integrating these perspectives is crucial, particularly in quantum computing, where traditional algorithmic limits are challenged. Due to the interdisciplinary nature of this topic, there is no unified concept that describes complex systems,[27] and the manner in which complexity can be measured quantitatively, rather than simply qualitatively, remains a subject of intense debate.

4. Quantum computing

4.1 Introduction to quantum information

In order to effectively implement a problem on a quantum computer, one must first grasp the basic principles of quantum systems and algorithms. Initially, it is worth addressing the question of what properties make quantum information so special. The concept of superposition is well-known in classical physics, for example regarding the linear combination of overlapping waves that results in constructive or destructive interference. However, in the quantum regime, it signifies the ability of a system to exist simultaneously in a linear combination of multiple states. Unlike bits, which encode a simple 0 or 1 state, qubits are represented mathematically by the state vector:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,. \tag{4.1}$$

Here, α and β are complex probability amplitudes, and their squared absolute values correspond to the probabilities of obtaining $|0\rangle$ or $|1\rangle$ state, satisfying the normalization condition:

$$|\alpha|^2 + |\beta|^2 = 1. \tag{4.2}$$

As shown in Fig. 4.1, the geometric representation of a qubit on the Bloch sphere is written as:

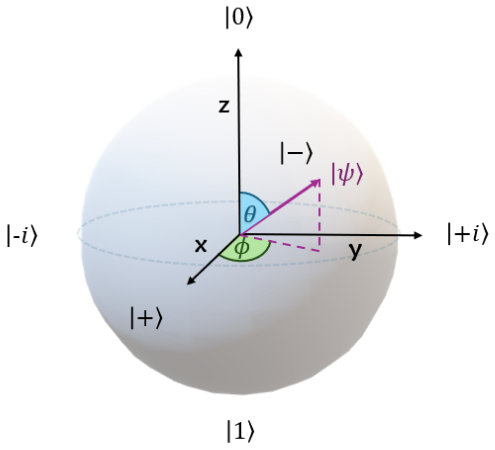


Figure 4.1: The Bloch sphere, providing a geometric representation of a single-qubit state.

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle\,. \tag{4.3}$$

The parametrs θ and ϕ are spherical coordinates. The polar angles describe probability amplitudes and azimuthal angle corresponds to the relative phase between basis state.

A system that connects all qubits and the operations (in unitary form) performed between them is called a quantum circuit.[28] These operators, usually referred to as quantum gates, are classified into different categories depending on their functionalities or the number of qubits on which they act. Several examples of these gates are summarized in Table 4.1.

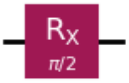
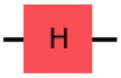
Name	**Circuit symbol**	**Matrix representation**	**Type of gate**	**Gate operation**
Pauli X		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	Single-qubit	Bit-flip
Rotation X		$\begin{bmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}$	Single-qubit	Rotation around X-axis
Hadamard		$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	Single-qubit	Creates superposition
Controlled Not		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	Multi-qubit	Entanglement / Conditional NOT
Arbitrary unitary operator			Multi-qubit	General state transformation

Table 4.1: Examples of quantum gates with their symbols and properties

A fundamental and intriguing phenomenon in quantum computing is quantum entanglement. Particles within such a system are intrinsically linked, meaning the state of one instantly depends on the state of the other, and they cannot be described independently. Experiments demonstrating this phenomenon through the violation of Bell inequalities were awarded the Nobel Prize in physics.[29] These results violate the assumptions of Bell's theorem regarding local realism (principle that states information cannot travel faster than the speed of light (locality) and that physical properties exist with definite values prior to measurement (realism)). Creating such an entangled system using qubits requires the application of a Hadamard gate and a CNOT gate, as illustrated in Fig. 4.2. An example of this is one of the Bell states:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \tag{4.4}$$

This state implies that while the first qubit has a 50% probability of being measured in state $|0\rangle$, a measurement result of 0 guarantees that the second qubit will also collapse into state $|0\rangle$. The same deterministic correlation applies to state $|1\rangle$. Such

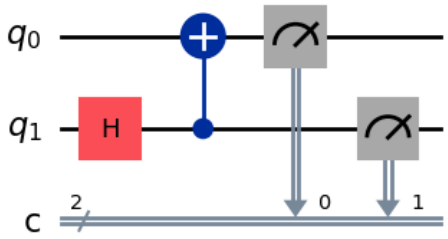


Figure 4.2: Quantum circuit for the generation of a Bell state $|\Phi^+\rangle$, containing 2 qubits and 2 classical bits for measuring the results.

a state cannot be represented on the Bloch sphere, as this representation is intended solely for pure states.

Although qubit measurement was briefly noted above, the underlying principles of how this process actually operates require further clarification. Extracting information regarding the state of a qubit requires mapping its quantum information onto a classical register. Within the Quantum Information Software Kit (Qiskit) framework, this process is executed[30] by projective measurements along the standard computational basis, which corresponds to the eigenstates of the single-qubit Pauli-Z operator. Consequently, the measurement operation projects the quantum state onto either $|0\rangle$ or $|1\rangle$, returning a classical 0 or 1. The likelihood of each outcome can be represented as:

$$|q\rangle \xrightarrow{\text{measure}} \begin{cases} 0 & (\text{eigenvalue } +1), \quad \text{with probability } p_0 = |\langle 0|q\rangle|^2 \\ 1 & (\text{eigenvalue } -1), \quad \text{with probability } p_1 = |\langle 1|q\rangle|^2 \end{cases} \tag{4.5}$$

4.2 Overview of quantum diagonalization algorithms

Algorithms that could potentially solve this electronic structure problem belong to the class of Quantum Diagonalization Algorithms (QDAs).[31] As their name suggests, their task is to solve matrix diagonalization problems,[24] which in this case reduces to determining the eigenvalues of the secular matrix (2.36).

4.2.1 Variational Quantum Eigensolver and Variational Quantum Deflation

Variational Quantum Eigensolver (VQE) is a fundamental algorithm that serves as the basis for many other, more complex variational frameworks. It features a division of labor between operations executed on a classical CPU and those performed on a QPU, as illustrated in diagram 4.3. The input requires a system-describing operator to be optimized (e.g., in the form of a Hamiltonian) and a trial state defined by a specific number of parameters tailored to the system. Although providing an initial trial state is not strictly mandatory, as initial parameters can be generated at random, the closer it is to the lowest-energy state, the faster the algorithm converges to the solution.

The primary objective of this hybrid algorithm is to determine the lowest eigenvalue based on the Rayleigh-Ritz principle:

$$\langle \Psi(\theta) | \hat{H} | \Psi(\theta) \rangle \geq E_0, \tag{4.6}$$

which states that the expectation value of any trial state will always be greater than or equal to the ground-state energy.

In this context, the ansatz is the parameterized quantum circuit that prepares the quantum state. The QPU is utilized to implement this circuit by rotating the qubits accordingly to prepere the trial state and then to measure and estimate the expectation value. Based on this result, a classical optimizer updates the parameters to minimize the cost function (the expectation value). Subsequently, a new variational quantum state is prepared using these updated parameters. This iterative process continues until the difference between consecutive estimated energy values ΔE falls below a tolerance ϵ or the maximum allowed number of iterations i_{max} is exceeded.

Using VQE alone is not enough, as the ground state is merely the beginning of a full solution; thus, the question arises whether it can be modified to achieve excited states. In fact, there is a closely related algorithm called Variational Quantum Deflation (VQD). This approach searches for the next lowest energies while excluding the ones that are already known. By adding a penalty term to the cost function, we prevent the algorithm from collapsing into the ground state each time, which yields:

$$\langle \Psi_1(\theta) | \hat{H} | \Psi_1(\theta) \rangle + \beta | \langle \Psi_0 | \Psi_1(\theta) \rangle |^2 \tag{4.7}$$

The inner product of independent functions vanishes because Hamiltonian eigenstates corresponding to distinct energies are orthogonal. If the states overlap, the value increases towards 1, thereby increasing the cost function due to the penalty parameter β.

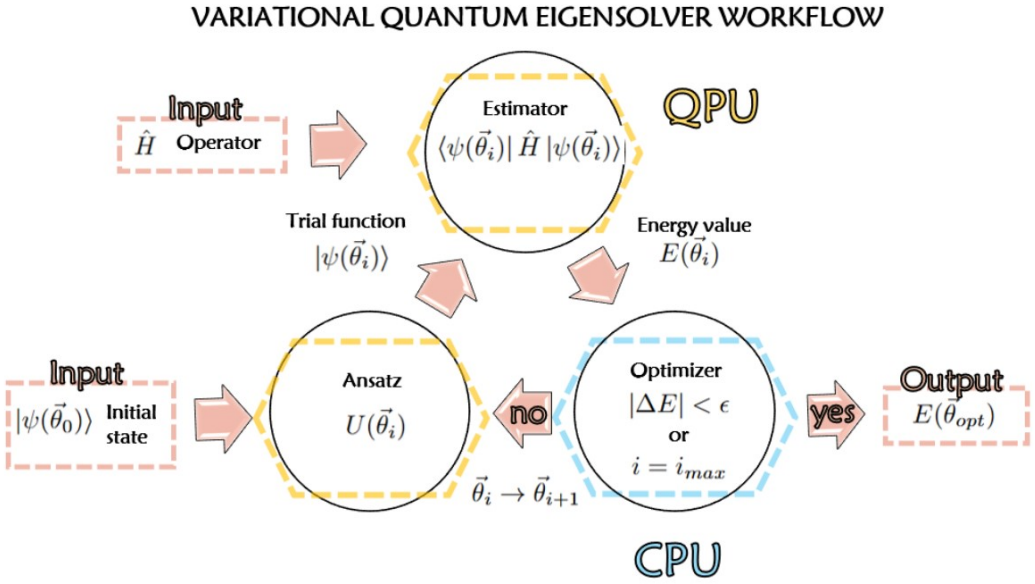


Figure 4.3: Diagram of the VQE alogithm, highlighting the input/oputput data and the division of tasks between the CPU and the QPU.

4.2.2 Other methods of quantum diagonalization

Subsequent hybrid algorithms have emerged, the first of which is Krylov Quantum Diagonalization (KQD).[32,33] Instead of the VQE approach based on parameter optimization, KQD generates states through time evolution and subsequently diagonalizes them classically. The Krylov subspace is constructed by repeatedly applying an operator associated with the Hamiltonian to a reference state $|\psi\rangle$, which can be expressed as:

$$\mathcal{K} = \{|\psi\rangle, e^{-i\hat{H}\Delta t}|\psi\rangle, e^{-i2\hat{H}\Delta t}|\psi\rangle, \ldots, e^{-i(r-1)\hat{H}\Delta t}|\psi\rangle\} \tag{4.8}$$

This process cannot be performed using a standard Hamiltonian, as it is non-unitary; therefore, it is realized via real-time evolution. Using the Hadamard test, the matrix elements and state overlaps are measured on the quantum processor, while the classical computer subsequently solves the generalized eigenvalue problem. On the other hand, the Sample-based quantum diagonalization (SQD)[34] algorithm generates subspaces through sampling, eliminating the need for time evolution. In this case, the CPU also processes the matrix data. Each of these three algorithms was developed for the (NISQ) era of quantum computing we are currently in, where full error correction is not yet possible due to imperfect hardware.

4.3 The Qiskit framework

4.3.1 Backends

All the necessary tools to create circuits, apply gates and execute them[28] can be found in Quantum Information Software Kit (Qiskit), a framework integrated with

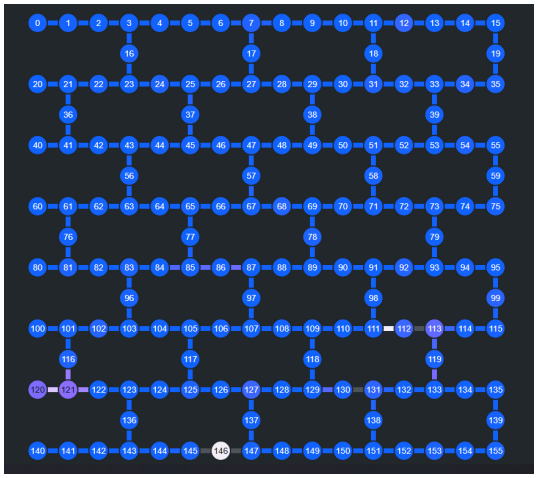


Figure 4.4: Qubit lattice of the ibm_kingston backend

IBM quantum hardware. Starting by discussing the types of backends that are utilized, IBM offers access to both ideal and noisy simulators of quantum computers alongside real quantum backends. Constructing a physical implementation of quantum computers is a widely studied topic. Quantum computers have been built based on trapped ions,[35] electronic spins in semiconductors (quantum dots),[36] nuclear magnetic resonance[37] or even topological qubits.[38] Those developed by IBM are constructed of transmon qubits. This type of qubit is made of two superconductors that exhibit zero electrical resistance below a critical temperature, separated by an insulating oxide barrier to form a Josephson junction. When no voltage is applied across the junction, a supercurrent flows, carried by electrons that form Cooper pairs. However, this type of qubit is not ideal. They are very sensitive to external noise like temperature fluctuations, leading to decoherence.

To illustrate quantum backend characteristics, one is analyzed as an example. The ibm_kingston backend, located in the us-east region, is one of the devices available for open use, meaning there is no need to pay for access. It is a Heron r2 type, which means it consists of 156 qubits. More qubits directly corelate to greater computational power. Qubits are designed in heavy-hexagonal lattice, shown in Fig. 4.4, this architecture reduces cross-talk between qubits, because each of them only has 2 or 3 neighbors. Another category is errors per layered gate (EPLG) that measures quality of qubits and quantum gates. The smaller this value, the better. For this backend 2Q error - the average two-qubit EPLG - is equal to 3.08×10^{-3}, For circuit layer operations per second (CLOPS), it is the other way around, more operations in fewer seconds yield a faster system. For this type of backend, its value is 340K. The time spent using real backends is limited, and access to certain versions requires payment.

Therefore, when large numbers of qubits are not required, either noisy or noiseless simulators can be utilized. It must be kept in mind that the amount of memory needed to describe a quantum state scales as 2^n. Despite these limitations, simulators remain a highly useful and constantly available tool, though for larger circuits, real backends must ultimately be used. Testing the selected program on a simulator beforehand is strongly recommended whenever possible, to avoid issues such as leaks

that would consume computational time.

4.3.2 Software tools and Qiskit modules

Expectation value estimation

The core architecture responsible for developing and executing quantum algorithms efficiently in Qiskit relies on Qiskit Primitives. Among them, the `Estimator` is a building block designed for the calculation of the expectation values of a given operator, also it is a necesery tool in VQD, which relies on iteratively evaluating expectation values. Qiskit's estimator algebraically partitions a complex observable into smaller components that can be determined simultaneously, which significantly minimizes the required measurement.[39,40]

Although commuting quantum operators can theoretically be measured at the same time, current technology only permits measuring operators that share a common tensor product eigenbasis. Readout on real hardware works on one qubit at a time, which can be performed along a single axis. During this procedure, the system breaks down into independent qubit outcomes, restricting evaluation to Pauli terms that share a common tensor product eigenbasis. It is equivalent to qubit-wise commutativity defined as a condition where each single-qubit Pauli operator in one product commutes with its counterpart in the other product. Finding the minimal number of such groups maps directly to the minimum clique cover, whicch is NP-hard.

In addition VQD need the state fidelity between the current trial state and previously found eigenstates. This overlap is typically measured with the `Sampler`[41] primitive, which ckecks the transition probability to the zero state.

Optimizers

To perform the optimization in VQD, it is necessary to use a classical optimizer. The Qiskit algorithms module contains various optimization algorithms,[42] which can be divided into local optimizers, looking for the best solution near a starting point, and global optimizers searching through all possible solutions. In this case, an algorithm that searches within a local neighborhood is required. For the most part, Constrained Optimization by Linear Approximations (COBYLA), represented by the `COBYLA` class, is utilized. It is a gradient-free (derivative-free) method, which makes it a popular choice for quantum computations because of its good resilience to statitical sampling noise, and it utilizes the COBYLA routine from `scipy.optimize.minimize`.[43]

Time evolution

To perform time evolution of a Hermitian operator H decomposed into Pauli terms, the following class is used `PauliEvolutionGate`.[44] This unitary operation is denoted by:

$$U(t) = e^{-itH}, \tag{4.9}$$

where t symolizes evolution real time. The evolution operator $U(t)$ corresponds to standard Pauli rotations up to a factor of two. Specifically, for the rotation around

the X-axis defined in exponential form as $R_X(\theta) = \exp\left(-i\frac{\theta}{2}X\right)$:

$$U(t) = e^{-itX} = R_X(2t). \tag{4.10}$$

This relation follows directly from equating the exponents:

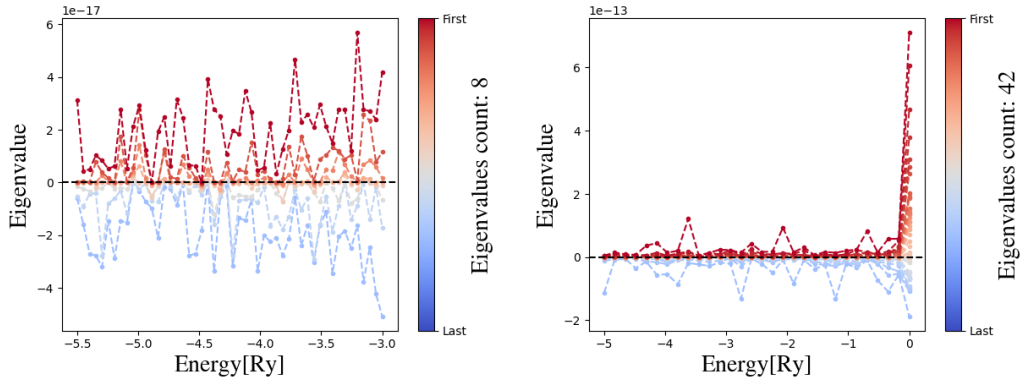
$$\frac{\theta}{2} = t \quad \Longrightarrow \quad \theta = 2t. \tag{4.11}$$

In practice, Qiskit does not construct $U(t)$ directly, but approximates it using different product formulas such as Trotterization (e.g., the default `LieTrotter` or the higher-order `SuzukiTrotter`), which can be controlled via the `synthesis` argument.

5. Quantum algorithmic approach to electronic structure theory

5.1 Form of the secular matrix

The secular matrix obtained in the equation 2.35 is a complex symmetric, but non-Hermitian matrix. This issue poses a huge challenge, because standard quantum algorithms, including VQD, are developed for Hermitian systems. Under such circumstances, two alternative paths are available: either adopting an algorithm for non-Hermitian matrices or transforming the matrix into a Hermitian form. Fortunately conventional calculations of the eigenvalues reveal that the magnitude of their imaginary part is negligible compared to the real parts calculated in the following section (Fig. 6.3). As illustrated in Fig. 5.1 the imaginary parts are of the order of 10^{-13} and 10^{-17}, while the corresponding real parts are on the order of 10^{3} and 10^{-1}. For the purpose of this study, this component is neglected, because it does not significantly affect the bound-state energies, which effectively resolves the issue of the matrix's lack of Hermiticity.



(a) (b)

Figure 5.1: Imaginary parts of eigenvalues for a system of (a) 4 and (b) 21 scatterers.

As previously mentioned, the VQD algorithm starts from the ground state and works its way up. Nevertheless, as will be shown later in the examples, the highest eigenvalues typically cross the axis, providing information about the bound state. To obtain the target results more quickly, the entire matrix is multiplied by -1, allowing to initiate the calculation from the more relevant side and avoid the accumulation

of numerical errors that worsen with higher excited states.

5.2 Decomposition of the secular matrix into Pauli terms

One of the most important tasks is mapping the secular matrix to an algorithm compatible format - Pauli string (`SparsePauliOp` in Qiskit). A relatively straightforward method[45] is used here to represent the operator in a format suitable for a quantum computer. In this approach, any complex square matrix representing the system's Hamiltonian or a general operator $A \in \mathbb{C}^{2^n \times 2^n}$ can be expressed as a weighted sum of Pauli strings:

$$A = \sum_{P \in \mathcal{P}_n} \alpha(P) P. \tag{5.1}$$

The summation runs over all possible elements belonging to the set:

$$\mathcal{P}_n := \left\{ \bigotimes_{j=0}^{n-1} P_j : P_j \in \mathcal{P} \right\}, \tag{5.2}$$

which is a set of n-qubit Pauli strings formed by taking the tensor product of n matrices chosen from the set of elementary Pauli operators $\mathcal{P} := \{I, X, Y, Z\}$:

$$I := \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad X := \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y := \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z := \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \tag{5.3}$$

Consequently, there are exactly 4^n terms in this sum. The coefficients $\alpha(P)$ are calculated by projecting A onto P using the trace operation:

$$\alpha(P) = \frac{1}{2^n} \mathrm{tr}(A^\dagger P) \quad (P \in \mathcal{P}_n). \tag{5.4}$$

5.3 Implementing VQD

Regarding VQD itself, there are a few aspects that should be taken into account. The penalty term in the cost function must be large enough to prevent falling back to the ground state, yet small enough in order not to exceed the energy gap to the excited states.

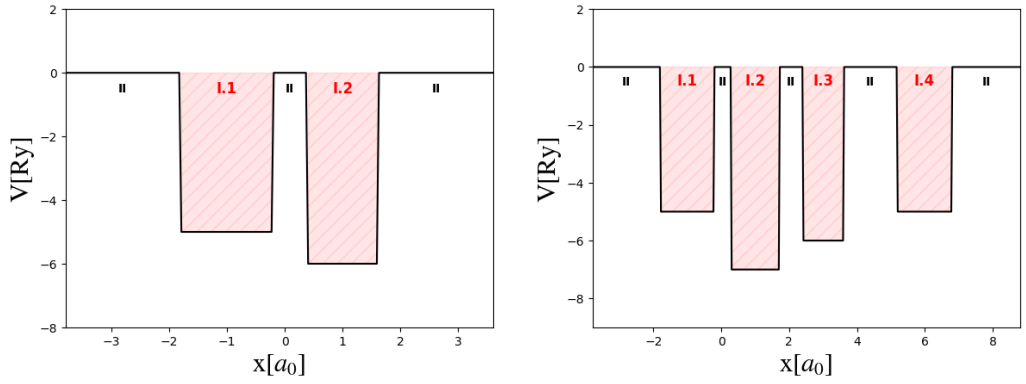
It is also worth noting that the structure and content of the code remains identical regardless of the fact whether calculations are performed on a simulator or a physical backend. Unfortunately, IBM changed their policy concerning the usage of real hardware. The free account on IBM's platform offers a 30-day trial with only a 10 minute runtime limit. Furthermore, the number of publicly accessible quantum processors has decreased, resulting in considerably longer queue times. It has limitations regarding queue priority when submitting jobs, and certain features are unavailable, such as the time-saving `qiskit_ibm_runtime.Session` mode.

Part II

Results

6. Computational results overview

To evaluate performance, computations are carried out for three distinct benchmark potential configurations consisting of 2, 4 and 21 scatters. Visualizations of these 1D systems are shown in 6.1(a), 6.1(b) and A, while their precise geometries are summarized in Table 6.1. Potential profiles are discretized onto a spatial grid. A margin of $\Delta x = 2a_0$ is added to both sides beyond the external scatterer positions. The grid resolution depends on the span between the minimal and maximal coordinates and is adjusted accordingly (e.g., for the smaller system in Fig. 6.1(a), the grid comprises 100 points). Although fewer points are needed to represent the potential wells, it becomes essential later to properly visualize the wavefunctions.



(a) (b)

Figure 6.1: One-dimensional potential profiles for configurations with (a) two and (b) four scatterers. The geometry of the two-scatterer setup is adopted from Butler[20] to validate the results obtained in this study against benchmark literature data. Dashed red regions indicate areas inside the potential, with each being assigned a label ($I.1$, $I.2$, etc.), whereas the interstitial regions are all designated as II. Following the form of Eq. (2.3), Rydberg atomic units are used, where the potential is given in Ry and the position in a_0.

6.1 Reference classical calculations

My first milestone is to construct routines that yield results using NumPy's built-in functions, relying on classical computation and the memory of a conventional computer. This library is capable of computing both determinants and eigenvalues,

Table 6.1: Summary of parameters including scatterer radii (r_i), well depths (V_i) and center positions (x_i), for each of the benchmark systems.

Configuration	$\mathbf{r_i}$ **(a_0)**	$\mathbf{V_i}$ **(Ry)**	$\mathbf{x_i}$ **(a_0)**
Two-scatterer	$\{0.8, 0.6\}$	$\{-5.0, -6.0\}$	$\{-1.0, 1.0\}$
Four-scatterer	$\{0.8, 0.7, 0.6, 0.8\}$	$\{-5.0, -7.0, -6.0, -5.0\}$	$\{-1.0, 1.0, 3.0, 6.0\}$
21-scatterer	0.15 (constant)	$\{-10.0 \cdots -1.0\}$ $(V_{\min}, V_{\max})$	$\{-9.5 \ldots 8.5\}$ $(x_{\min}, x_{\max})$

and I utilize both of these functionalities to calculate properties of the secular matrix Eq. (2.36). The determinants are examined solely to verify the eigenvalue results, since the eigenvalues are the center of attention, given that they are the quantities we can compute using quantum algorithms. The energy points can be selected from the interval $(V_{min}, 0)$, where V_{min} corresponds to the global minimum of the potential. However, values too close to these boundaries should be avoided to prevent numerical instabilities.

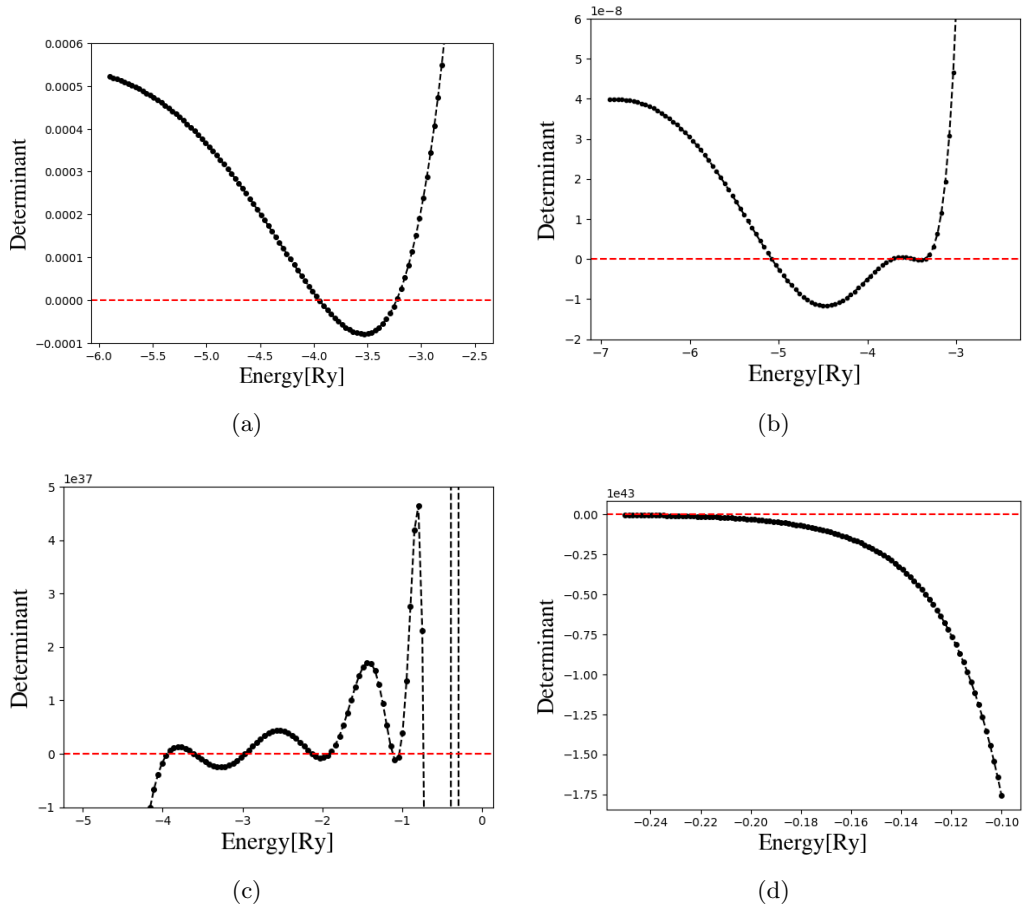


Figure 6.2: Close-up views of the energy-dependent secular determinant for systems with different numbers of scatterers: (a) two scatterers, (b) four scatterers, (c) 21 scatterers, and (d) again for 21 scatterers, illustrating the numerical explosion.

Fig. 6.2 shows selected fragments of the determinant graphs for the investigated systems. In particular, the range of each plot is adjusted primarily along the y-axis to enhance visibility. As predicted, the number of x-axis crossings and consequently the number of allowed energy states, increases with the system size.

Evidently, the number of eigenvalues is twice the number of scatterers (Fig. 6.3). Furthermore, a tendency for the highest eigenvalues to cross the axis can be observed, which suggests that the most efficient approach in subsequent computational stages is to begin with them. Most eigenvalues decrease monotonically with growing value of energy, but as they approach zero their behavior becomes more unpredictable.

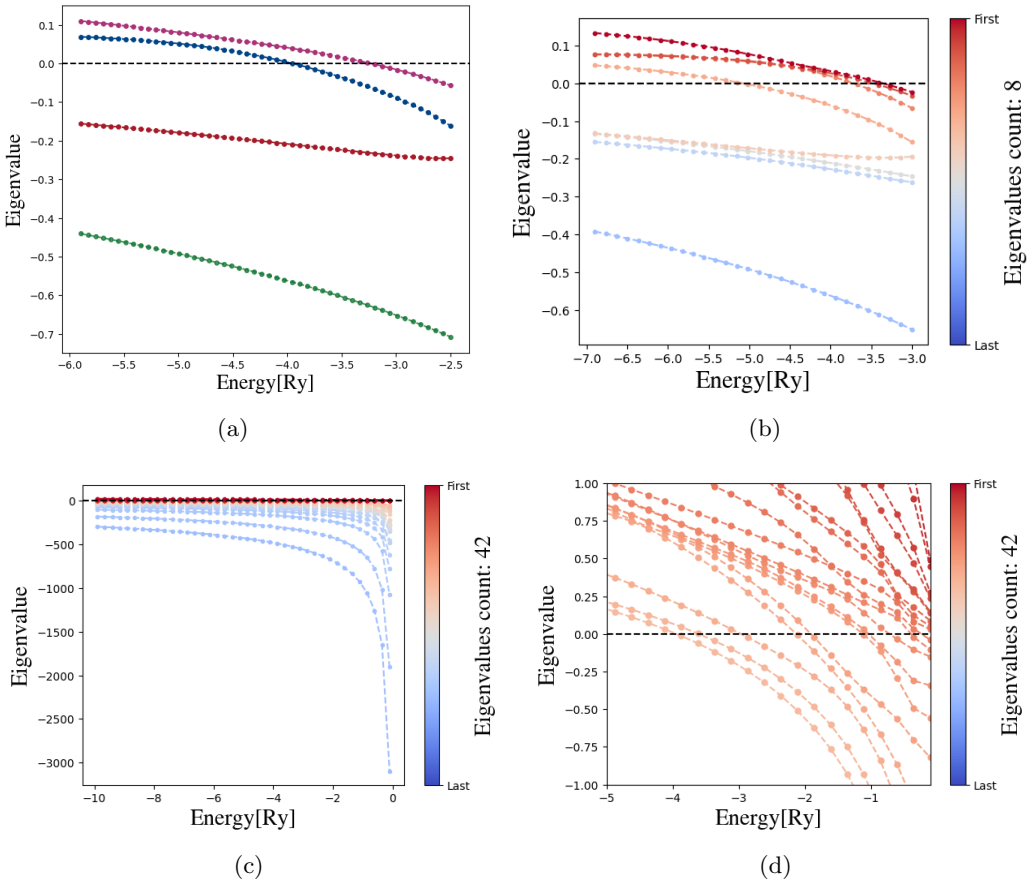


Figure 6.3: Eigenvalue spectra as a function of energy for varying system sizes: (a) two scatterers, (b) four scatterers, (c) 21 scatterers and (d) corresponding close-up view. Additionally, for systems with more than two scatterers, the plots feature a color gradient distinguishing individual eigenvalue, along with the total number of eigenvalues displayed on the side.

Although tracking individual eigenvalues might initially appear more computationally demanding than evaluating the secular determinant, for more complicated systems it develops extremely steep gradients, leading to potential overflow issues. Indeed, for only 21 scatterers, its values escalate to approximately 10^{43}. Tracking individual eigenvalues is therefore far more stable and reliable, given the considerably smoother nature of their curves.

Table 6.2: Interpolated allowed energies, along with the adjacent energy points bracketing each sign change identified for the 21-scatterer configuration.

Eigenvalue number	**In between the points**	**Interpolated value [Ry]**
0	$[-4.00, -3.90]$	-3.944302
1	$[-3.70, -3.60]$	-3.608752
2	$[-3.00, -2.90]$	-2.968744
3	$[-2.20, -2.10]$	-2.127734
4	$[-2.00, -1.90]$	-1.907422
5	$[-1.20, -1.10]$	-1.116719
6	$[-1.10, -1.00]$	-1.028155
7	$[-0.80, -0.70]$	-0.724051
8	$[-0.40, -0.30]$	-0.379377
9	$[-0.30, -0.20]$	-0.273111

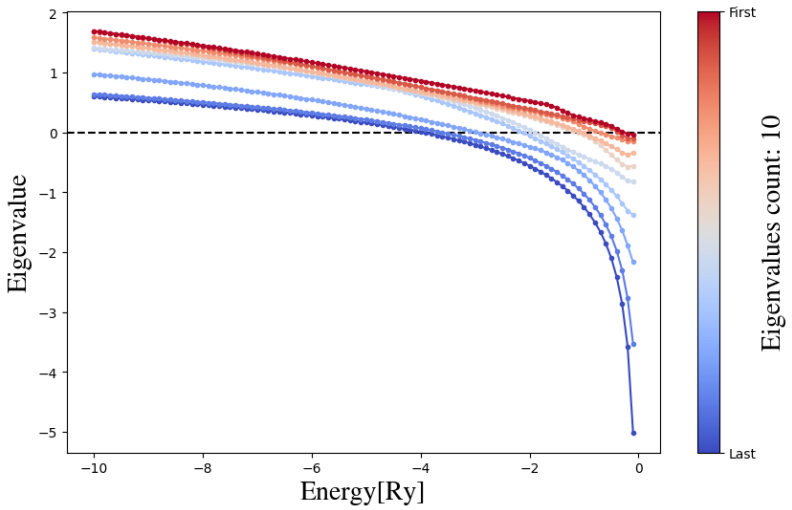


Figure 6.4: Plot showing only the eigenvalues selected for polynomial interpolation for the 21-scatterer system.

However, such an approach only allows for a very rough estimate of the value closest to zero, or simply identifies the interval in which the zero-crossing occurs. Limiting the results to the discrete points calculated within the given range is insufficient. Therefore, an interpolation technique has been developed, in which the four nearest neighboring points are selected and fitted with a third-degree polynomial. This is demonstrated for the 21-scatterer system by discretizing the energy range $(-9.1, 0.1)$ into 100 points, identifying which eigenvalues exhibit a sign change in this interval (in this case, 10 eigenvalues shown in Fig. 6.4), and subsequently locating and approximating these crossing points to yield the 10 energy values (Tab. 6.2).

6.2 Comparison of VQD results

Following the successful implementation of the MST formalism, confirmed by the expected behavior of the eigenvalue plots, and the development of an interpolation pipeline to extract energy states, the focus shifted to the key objective: employing Qiskit tools to perform calculations using the noiseless `AerSimulator`,[46] with execution on quantum hardware reserved as a next step. The implementation strategy relies on the selection of an energy range and resolution over which the VQD algorithm will be executed. Because each energy value yields a distinct secular matrix, a separate run of the VQD is required each time. The objective is to minimize the number of sampling points while maintaining satisfactory accuracy. Before doing so, it is necessary to first outline the essential transformations required to pass this matrix to the algorithm.

Table 6.3: Quantum resource requirements: secular matrix dimensions, qubit counts and Pauli term scaling across benchmark scatterer configurations.

Number of scatterers (n)	**Secular matrix size**	**Required qubits (N_q)**	**Max Pauli terms (4^{N_q})**	**Non-zero Pauli terms**
n ($n \in \mathbb{Z}^+$)	$2n \times 2n$	$2n \leq 2^{N_q}$	4^{N_q}	$\leq 4^{N_q}$
2	4×4	2	16	3
4	8×8	3	64	14
21	42×42	6	4096	944

For all three study cases, I created a test secular matrix for an energy of -3 Ry to check if the number of Pauli terms corresponds to the previously mentioned 4^n, where n is the qubit number. Table 6.3, presents a comparison of the number of scattering centers, the dimensions of the secular matrix, the number of required qubits, the maximum number of Pauli terms and their non-zero elements. All results are in agreement with the predictions from section 5.2.

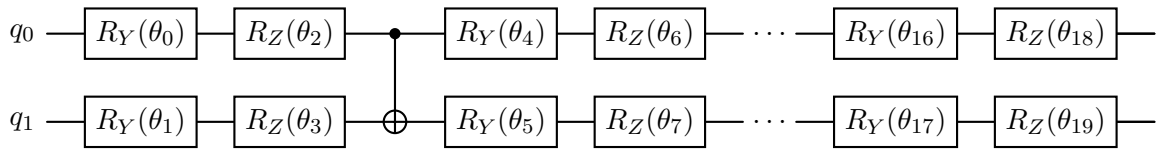
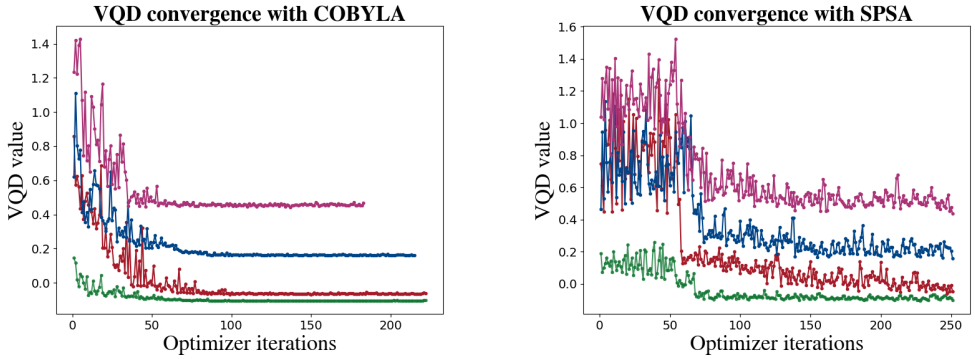


Figure 6.5: Schematic representation of the parameterized `EfficientSU2` ansatz.

To proceed with the implementation of the VQD algorithm, several additional components must be prepared beyond the decomposition of the matrix into Pauli operators. The type of ansatz used is called `EfficientSU2`,[47] which is very commonly used in variational quantum algorithms. Based on the example circuit layout shown in Fig. 6.5, the internal structure of the ansatz can be analyzed in detail. It is composed of parameterized single-qubit rotation gates belonging to the special unitary group SU(2), specifically rotations such as $R_Y(\theta)$, $R_Z(\theta)$, or $R_X(\theta)$. The subsequent rotation layers are interconnected through two-qubit entanglement op-

erations, implemented here using Controlled Not gates. This architecture belongs to the class of hardware-efficient ansätze, designed to match a near-term quantum processor. The ansatz must be initialized with parameters from the $[0, 2\pi]$ interval.

Each individual value calculation involves multiple iterations, following the previously mentioned variational principle (Eq. (4.6)) that drives the optimization of the VQD parameters. Figures 6.6 show how two optimizers achieve convergence for individual eigenvalues in the case of two scatterers. A crucial aspect involves choosing an optimizer fitted for a given problem; some may achieve convergence in too many iterations (e.g., 6.6(b)), while others may give completely wrong results. Consequently, COBYLA is chosen as the most suitable solution.



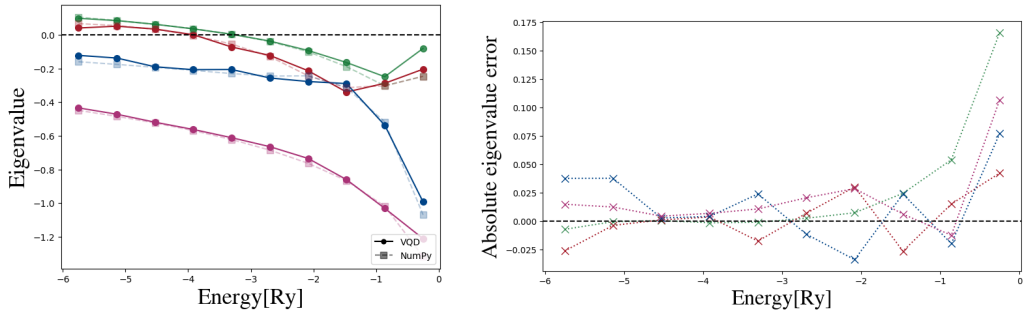
(a) (b)

Figure 6.6: Optimization process across a single VQD run for the first four eigenvalues using different Qiskit optimizers for the system with two scatterers.

Table 6.4: Comparison of calculated energy levels across Butler's calucations,[48] classical and quantum methods for the two scatterer benchmark system.

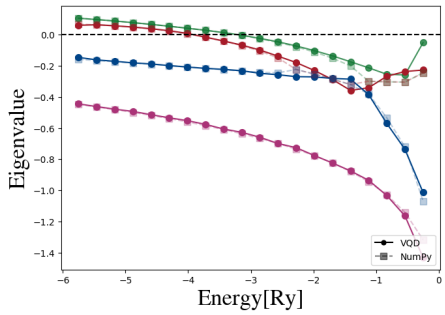
Bound state energy	**Numpy** (22 energy points)	**VQD simulator** (22 energy points)	**Butler**
E_0	-3.949869	-3.959828	-3.949902
E_1	-3.228485	-3.205805	-3.228487

The algorithm is first applied to the initial benchmark system composed of two scatterers. Values are evaluated across 4 to 40 energy points in steps of two, with all ansatz parameters initialized to π for each run. Selected plots, along with a comparison with NumPy baseline calculations, are shown in Fig. 6.7. These results show, for the most part, agreement with classically obtained calculations. However, occasionally numerical deviations can be observed which typically occur more frequently for eigenvalues computed later in the VQD process. It is also worth noting that the errors grow significantly as eigenvalues near zero. This is primarily caused by a lack of monotonicity and the presence of bifurcations, which make it difficult for VQD to distinguish between adjacent states. A comparison of the bound-state energies

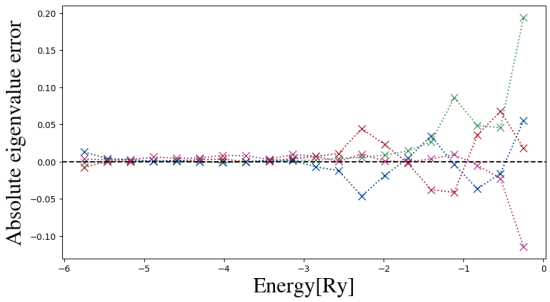


(a) 10 points

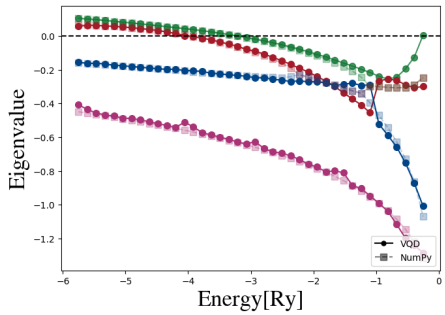
(b) 10 points



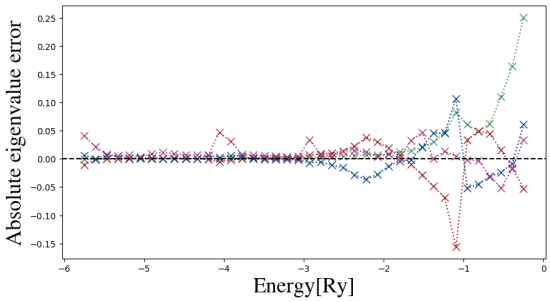
(c) 20 points



(d) 20 points



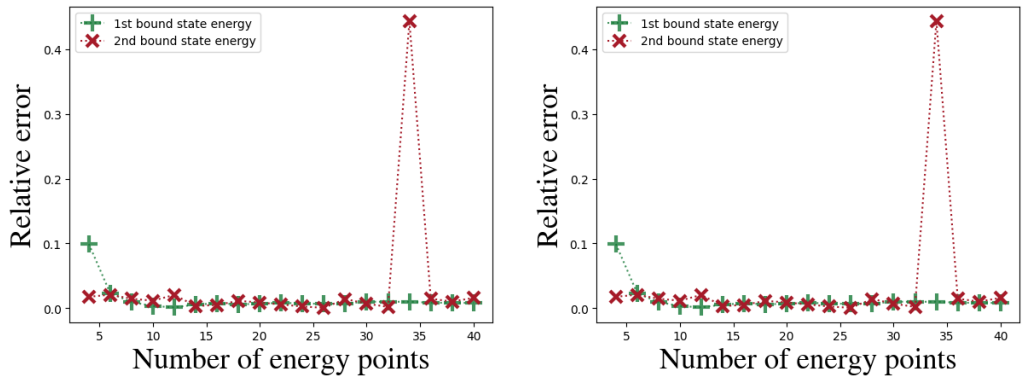
(e) 40 points



(f) 40 points

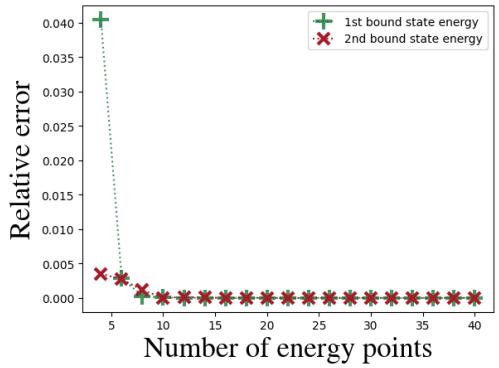
Figure 6.7: Comparison of eigenvalues computed using VQD and NumPy for the two-scatterer system at a chosen energy resolution. Alongside the eigenvalue comparison, the absolute error relative to the NumPy calculations is displayed.

interpolated from the computed eigenvalues is presented in Fig. 6.8. The computational error drops sharply only up to 10-12 energy points. Beyond this threshold, the NumPy error plateaus, but VQD error oscillates. At 34 points a huge spike in error occurs as a consequence of a large inaccuracy in the eigenvalue evaluation, resulting in an artificial intercept with the energy axis. Tab. 6.4 presents the summary for 22 energy sampling points compared with those reported in the Butler's paper.[48]

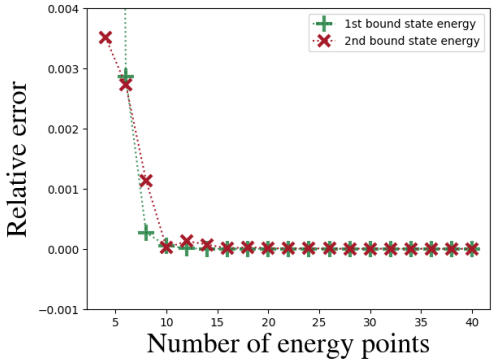


(a) VQD

(b) VQD

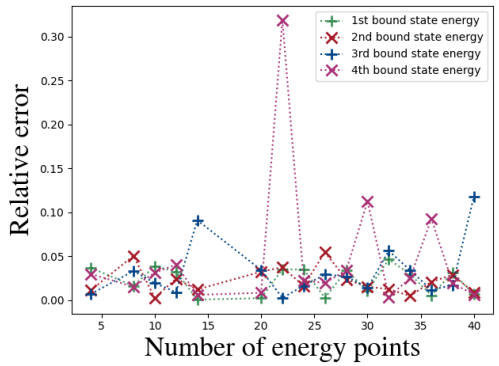


(c) NumPy

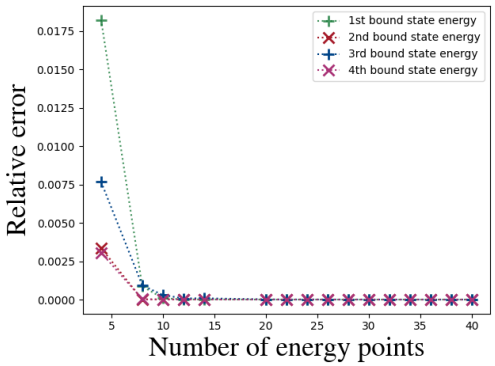


(d) NumPy

Figure 6.8: Relative error of the interpolated bound states obtained by VQD and NumPy (with respect to high-accuracy reference values computed by NumPy for $1,000$ energy points) as a function of the number of energy points used in calculation for the two-scatterer system, showing both the full-scale plots and magnified views.



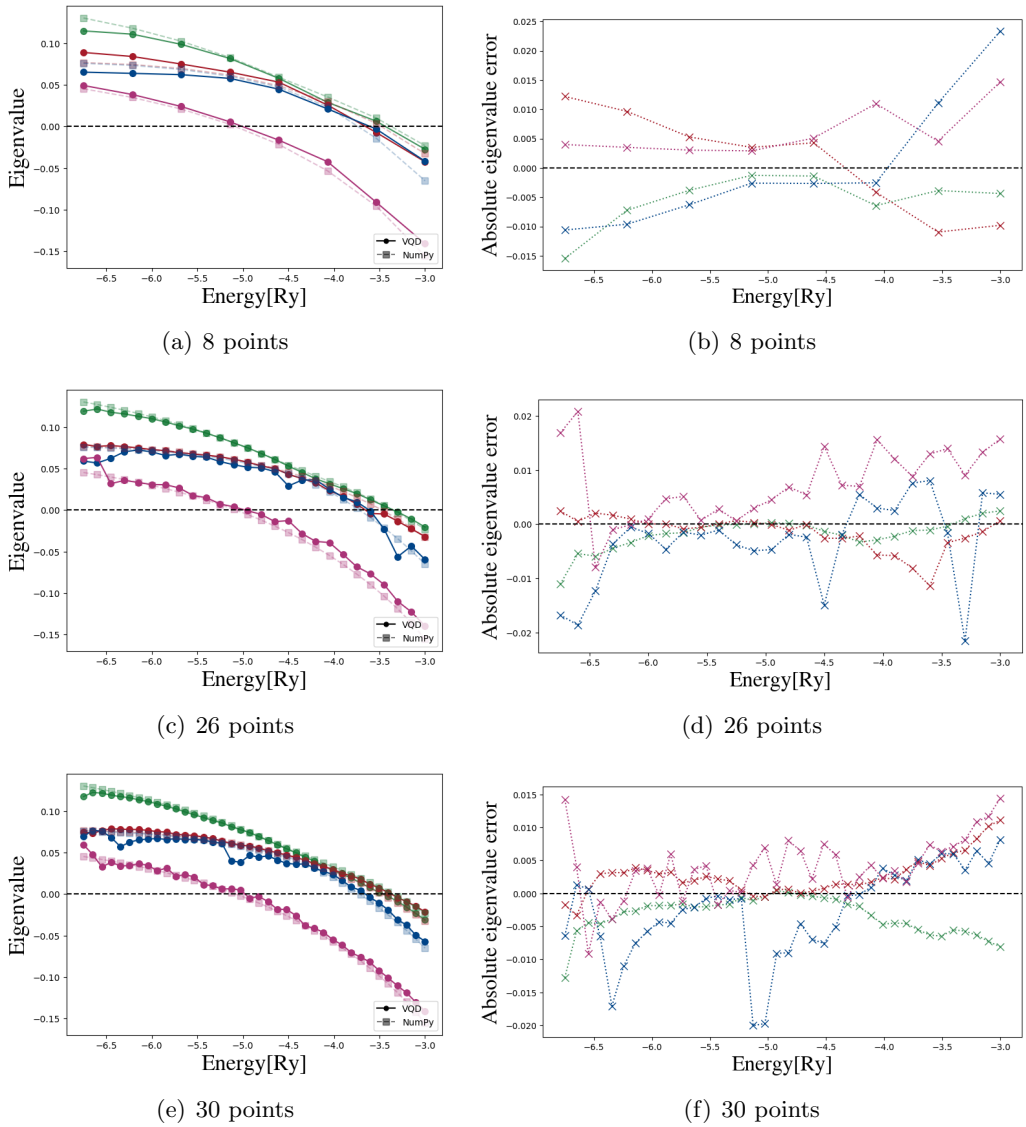
(a) VQD



(b) NumPy

Figure 6.9: Relative error of the interpolated bound states obtained via VQD and NumPy, computed relative to high-accuracy reference values computed by NumPy for $1,000$ energy points, versus the number of energy points used in calculation for the four-scatterer system.

The same process is applied to the system of four scatterers (Fig 6.10). Even for four scatterers, it is necessary to introduce an iteration limit for the optimizer to obtain results within a reasonable timeframe. This case is a much greater challenge than the previous one, particularly because the first three eigenvalues are very close to each other, making this task especially difficult for VQD. That is why despite the monotonicity of these segments, numerical errors still occur. Just as in the smaller system, occasional significant deviations can be observed. Nevertheless, the majority of the data points closely follow the NumPy curve. The interpolation of the bound states (Fig. 6.9) is considerably less accurate. There are no clear trends when increasing the resolution, largely due to the unpredictable nature of the calculations caused by the close proximity of the eigenvalues.



(a) 8 points (b) 8 points

(c) 26 points (d) 26 points

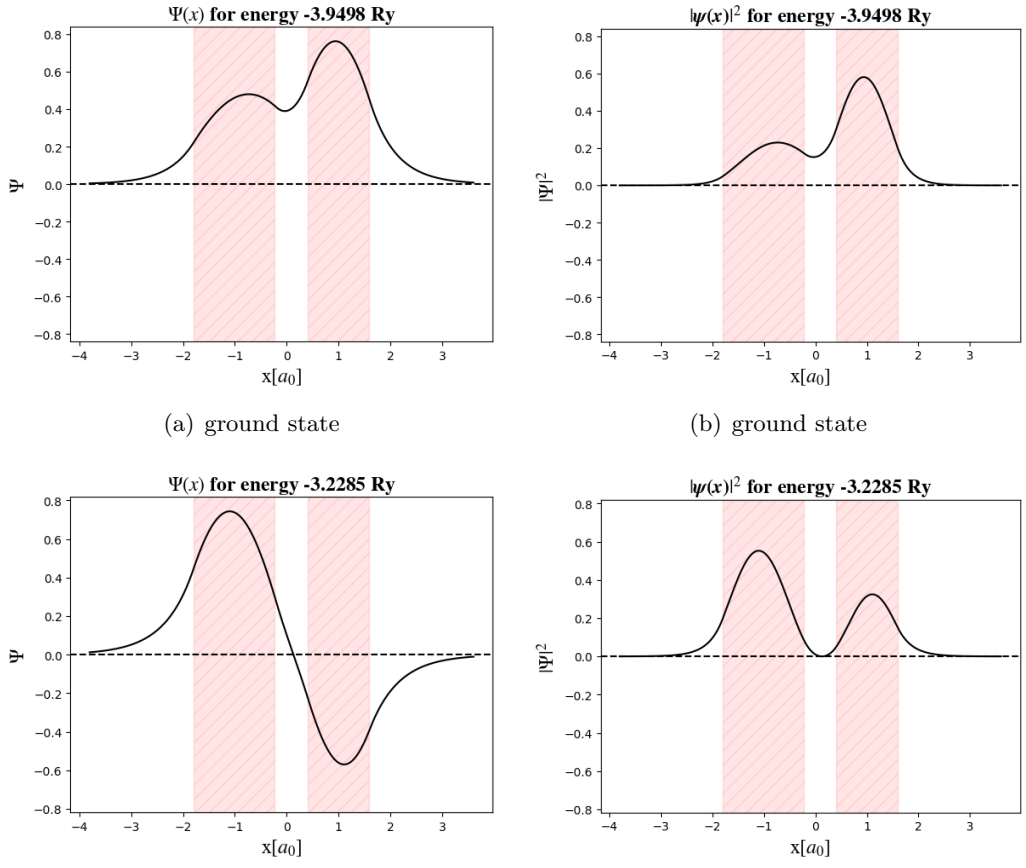
(e) 30 points (f) 30 points

Figure 6.10: Comparison between VQD and NumPy eigenvalues for the four-scatterer system at a given energy resolution. The absolute error relative to the NumPy reference values is also shown.

Since only the eigenvalues beyond the tenth cross the zero axis for the 21-scatterer system and calculating higher eigenvalues yield progressively less accurate results with VQD, no presentable results could be obtained. Additionally, similarly to the four-scatterer case the eigenvalues are located very close to one another.

Unfortunately, it is also not possible to reproduce these results on real quantum backends. Due to recent changes in IBM's policy, access is considerably restricted. The free-tier plan does not provide enough resources to obtain results even for the two-scatterer system.

6.3 Results verification - plotting wavefunctions



(a) ground state

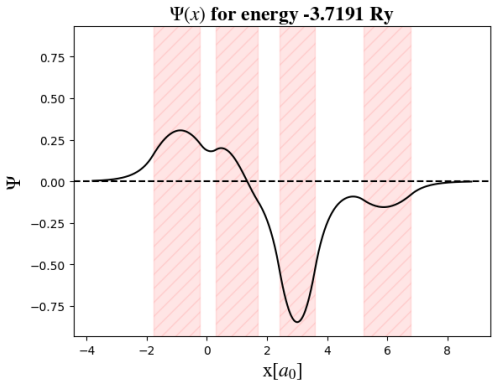
(b) ground state

(c) first excited state

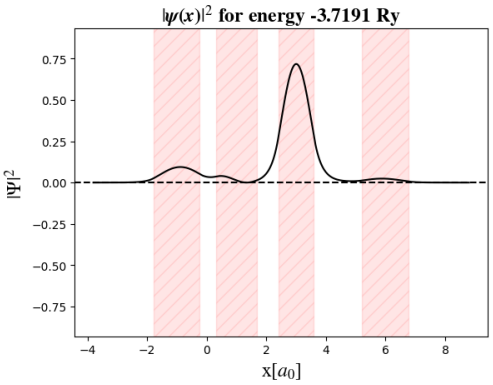
(d) first excited state

Figure 6.11: Wavefunctions and their squared moduli for bound-state energies of the benchmark two-scatterer system.

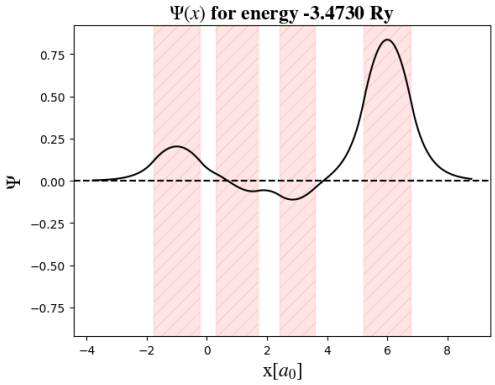
An excellent method for verifying the calculated bound state energies is plotting the wavefunctions corresponding to the obtained results. The method for obtaining the corresponding coefficients is described in Section 2.2.3 and based on this approach the procedure for calculating these functions is implemented. Wavefunctions are normalized using Simpson's rule integration. The plots for the benchmark systems are presented in Figures 6.11, 6.12 and 6.13. As in the case of the potential



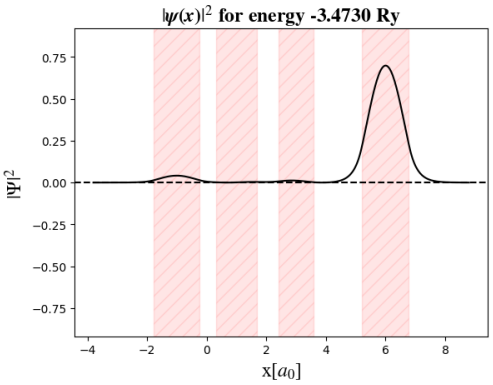
(a) first excited state



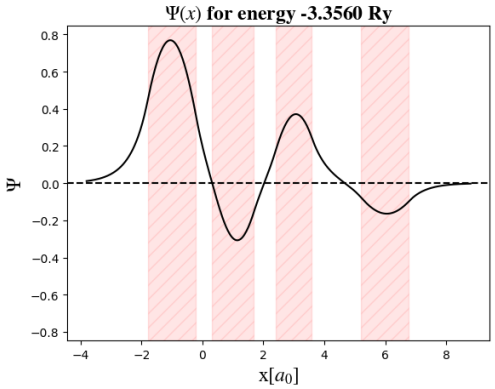
(b) first excited state



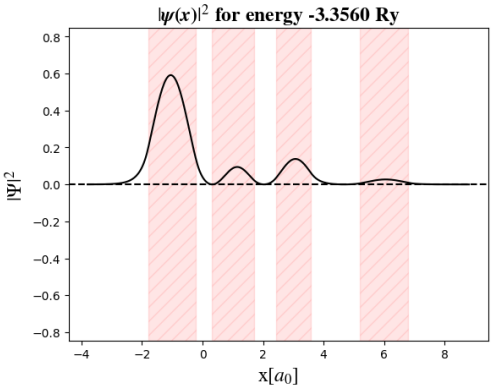
(c) second excited state



(d) second excited state



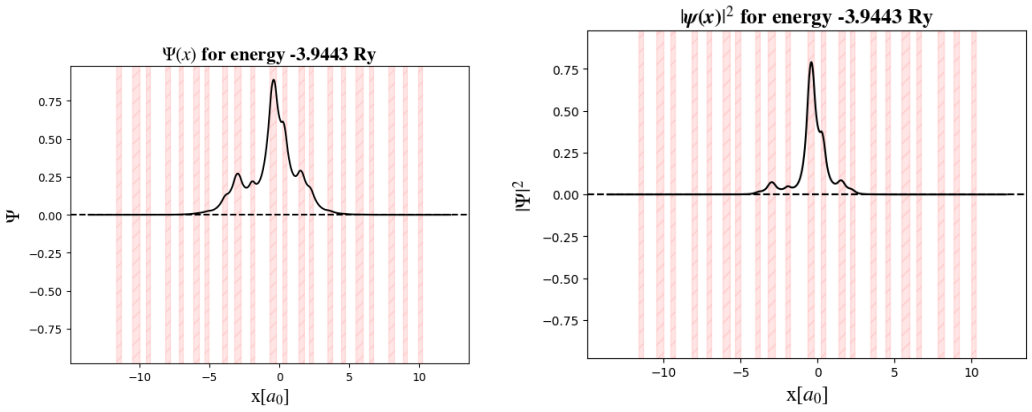
(e) third excited state



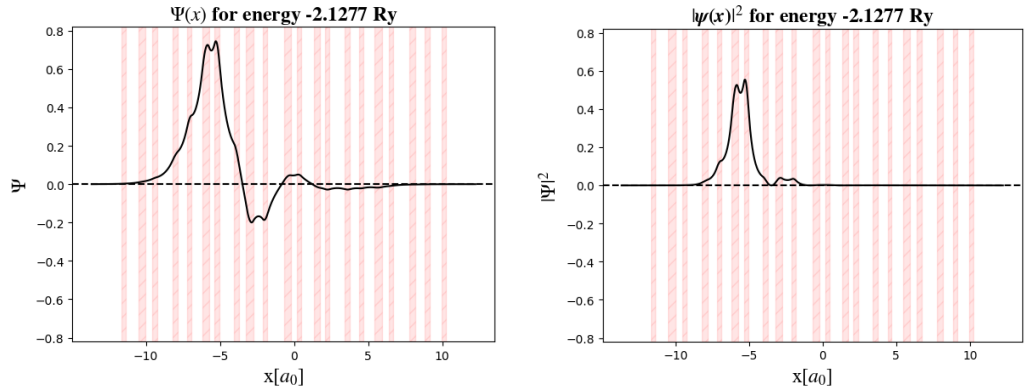
(f) third excited state

Figure 6.12: Wavefunctions and their squared moduli for selected bound-state energies of the benchmark four-scatterer system.

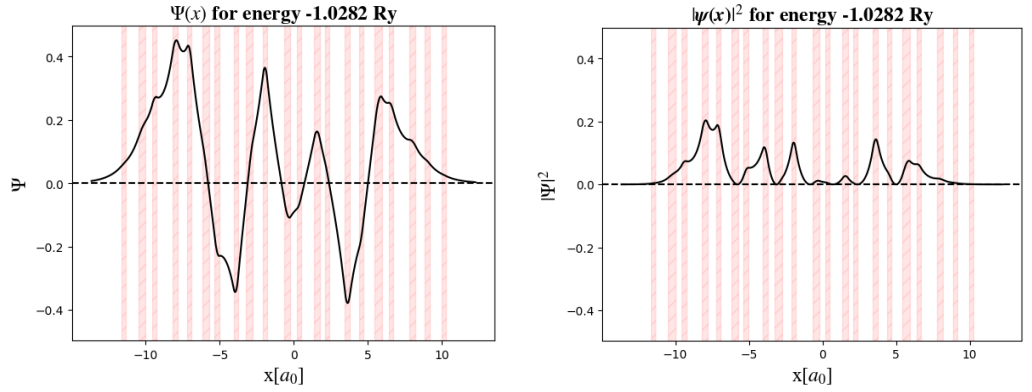
profile, the red dashed regions denote the interiors of the scatterers. As can be seen, for the correct energies, wavefunctions are continuous and each segment smoothly transitions into the next. Furthermore, the plots for the two-scatterer case match those reported in Butler's publication.[48] To the right of the wavefunction plots,



(a) ground state (b) ground state



(c) third excited state (d) third excited state



(e) 6th excited state (f) 6th excited state

Figure 6.13: Selected wavefunctions and their squared moduli for bound-state energies of the benchmark 21-scatterer system.

squared moduli of the wavefunctions are shown, from which the charge density can be calculated. Moreover, when extending the formalism to 2D or 3D, there would be a possibility for the spatial charge distribution to be computed analogously.

7. Novel quantum algorithm for MST problem solution

Besides using previously developed algorithms such as VQD, this work aims to develop an approach that does not require as many iterations and quantum circuit executions. VQD provides an excess of information that is not strictly necessary, leading to a significant waste of resources, such as time and utilization of quantum hardware, without generating any new insights. To ensure that the computational cycle is optimized, the required computational capacity must be minimized.

Instead of calculating a set of secular matrices for a fixed set of energy points, a more flexible approach is proposed, focusing only on the values of interest. Instead of running the same algorithm multiple times, what about adapting it to also optimize energy. One way to achieve this is by incorporating energy directly as a parameter into the ansatz and then including it some way in the cost function. With every newly optimized energy provided by the algorithm, a corresponding secular matrix is generated without wasting time on unnecessary computation. However, this brings up another issue to consider, namely the structure of the resulting cost function and how to ensure that the energy converges to a bound state.

An important point to note is that a properly executed VQD should result in obtaining an eigenstate of the matrix acting as an operator in this algorithm. Consequently, the time evolution of the resulting ansatz should not alter its form (apart from a global phase factor). Every eigenvalue corresponds precisely to such a stationary state. With this in mind, this knowledge can be utilized to reward states that are approaching the eigenstates of the Hamiltonian, while simultaneously driving the value toward zero, as this directly guides the optimization toward the bound-state energy. For this reason, a time evolution is performed for an ansatz and then evalute how much the state changes using Hellinger distance, which compares the measurement probability distributions of the two states (`hellinger_distance`).[49]

The goal is to start with a random point and parameters, approach any eigenvalue, and then follow these values until the intersection with the energy axis occurs. The workflow of this algorithm is structured as follows:

1. Random parameters are generated from 0 to 2π and an initial energy value is appended at the end of the parameter vector (again in the range from the deepest well to 0, but values closer to zero are selected to provide a clearer visualization of how the algorithm works).

2. An ansatz is prepared with the appropriate number of qubits and a classical optimizer is selected.

3. Inside the cost function, the following steps are performed:

(a) The parameters, ansatz, β and t are passed as inputs, where t is the state evolution time and β represents the weight of the segment associated with the Hellinger distance.

(b) If the energy falls outside the range from the maximum depth of the well to 0, the function returns a large penalty value to avoid further exploration of this region.

(c) If the energy is within the valid range, the secular matrix for the given energy is constructed, multiplied by -1 (the additive inverse), converted into a Pauli operator, and the expectation value for the current ansatz, parameters and operator is estimated.

(d) The resulting state is then copied and a Pauli time evolution is performed for the duration t. The counts of both states are then measured and the Hellinger distance is calculated to quantify their divergence.

(e) Finally, the total cost function is constructed as follows:

$$\text{cost} = |\text{eigenvalue}| + \beta \times \text{dist}_{\text{hellinger}} \tag{7.1}$$

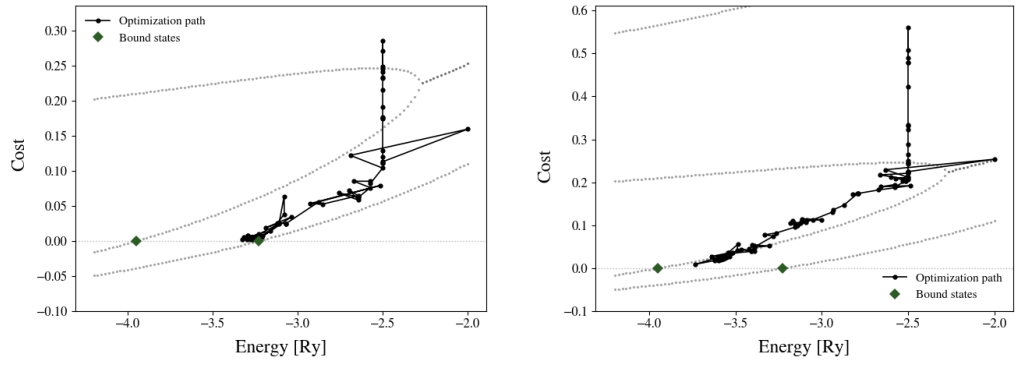
4. The classical optimizer updates the parameters iteratively until convergence is reached.

In summary, the primary difference between this algorithm and the standard VQE lies in the use of the absolute value of the eigenvalue, as the objective is to get as close to zero as possible. Unlike VQD, where running the algorithm for multiple energy points yields a set of eigenvalues used for estimation, here intermediate points along the optimization trajectory do not directly correspond to valid eigenvalues. The cost function contains a term associated with the Hamiltonian time evolution and the Hellinger distance, which evaluates how close a given state is to being an eigenstate and penalizes states that fail to show stationary behavior.

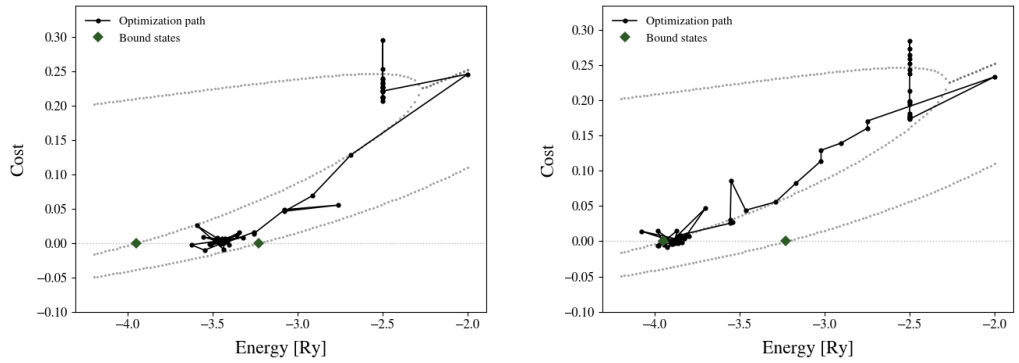
As illustrated in the figures 7.1 showing the estimation of the allowed energies, the algorithm exhibits a tendency to move towards the lowest possible values. But once it encounters regions near the eigenvalues (indicated by the gray points), it begins to track them. Despite fulfilling these core principles, the final results do not consistently align with the expected values. For example, in Figure 7.1(b), where the optimization ends while tracing an eigenvalue curve that does not intersect zero. Furthermore, some runs yield inaccurate results despite coming close to the energy axis intercept, while others fail entirely by converging to insignificant regions.

It is hypothesized that these errors might be due to the choice of parameters t and β. To investigate this, numerous runs of the algorithm are performed to generate heat maps (Fig. 7.2), where the parameters serve as the axes, and the intensity represents the relative error between the value obtained by the algorithm and the reference energy. However, no clear trend is observed in these results.

Although the proposed algorithm does not yet produce consistently accurate results, it represents an intriguing and considerably faster alternative to existing approaches. The continued work on its optimization and the design of more effective cost functions remains a promising direction for future research. Notably, the algorithm is capable of converging toward any arbitrary bound state. Once fully developed, it would be advisable to introduce a penalty term analogous to VQD to penalize convergence toward previously identified states.

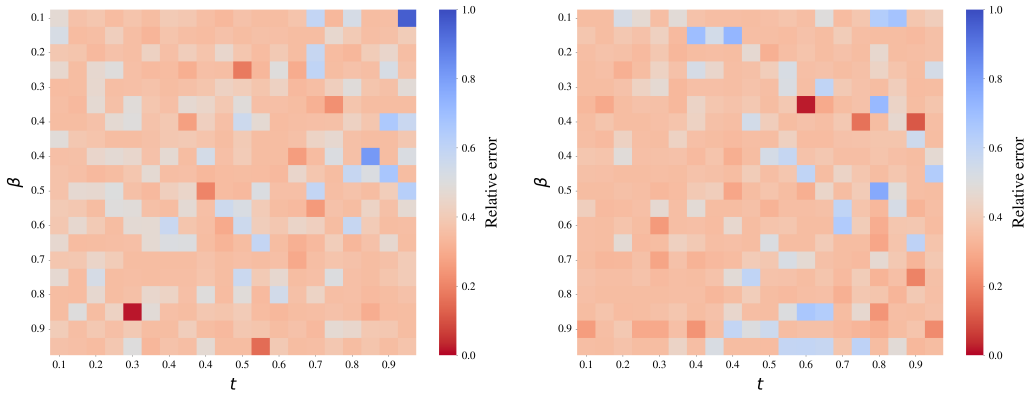


(a) (b)



(c) (d)

Figure 7.1: Optimization trajectories of the proposed algorithm attempting to find the bound state energy for parameters $t = 0.1$ and $\beta = 0.5$, where (a) convergence to the first excited state, (b) descent toward the ground state that stops prematurely, (c) a failed run converging to an irrelevant value, and (d) convergence to the ground state.



(a) (b)

Figure 7.2: Example heatmaps showing the accuracy of the custom algorithm's results when run with various β and t parameters.

8. Scalability and complexity gain analysis

8.1 Time and space resource consumption in NumPy diagonalization

A crucial aspect is evaluating the computational efficiency of all the approaches used to calculate the eigenvalues of a square matrix. First, the `eigvals`[50] function from NumPy's linear algebra module `numpy.linalg` is considered. It relies on finite precision arithmetic and routines from the LAPACK[51] library. Specifically, this function utilizes the _geev driver, which depending on the input data type and precisions, has a few subroutines. These procedures do not employ ultra-fast algorithms, but rather a standard classical method. Consequently, it is assumed that:

$$\omega = 3 \implies T_{MM}(n) = \mathcal{O}(n^3) \tag{8.1}$$

In practice, the parameter δ factor is fixed by the precision of the machine, which causes $\mathcal{O}(\log^2(n/\delta))$ to be reduced to $\mathcal{O}(1)$. For this reason, it is assumed that the computational complexity of this procedure scales as $\mathcal{O}(n^3)$.

Nevertheless, it is worth verifying how the problem actually scales in real computation, rather than relying solely on theoretical assumptions. The time required by a standard consumer computer to compute the eigenvalues of increasingly large matrices is evaluated (Fig. 8.1). Subsequently, a trend line ($t \propto n^3$) is fitted to the obtained data, which aligns with previous predictions.

The same procedure is employed for memory consumption (Fig. 8.1). During the eigenvalue computation process, RAM usage is monitored every 50 ms, whenever a higher value is recorded, it replaces the previous peak. Once the maximum memory usage for a given matrix is determined, the baseline consumption measured before the start of the procedure is subtracted. This process is repeated for matrices of various sizes. However, in this case, an $\mathcal{O}(n^3)$ scaling is not assumed, as the tracked resource is not the number of executed operations, but the allocated memory. Instead, a $\propto n^2$ trend line is fitted because LAPACK operates directly on the matrix and creates a constant number of copies.

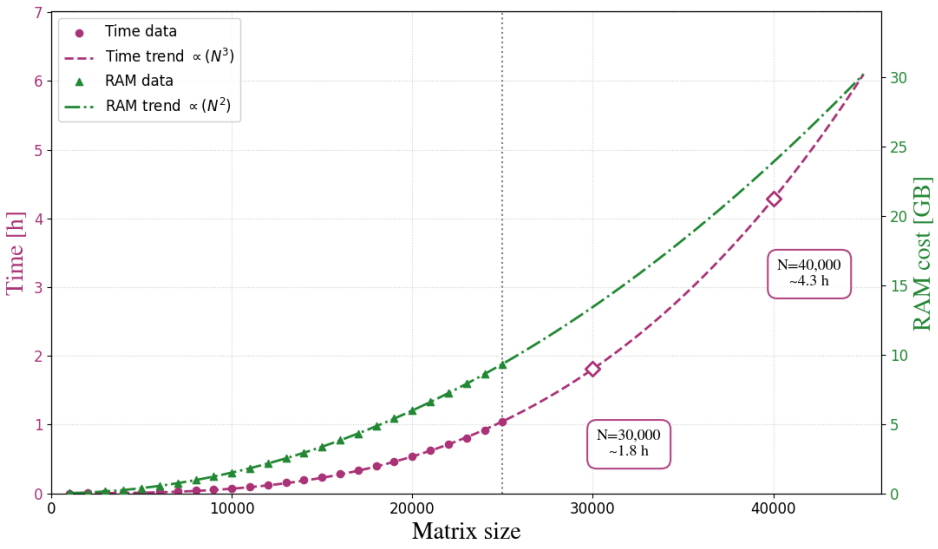
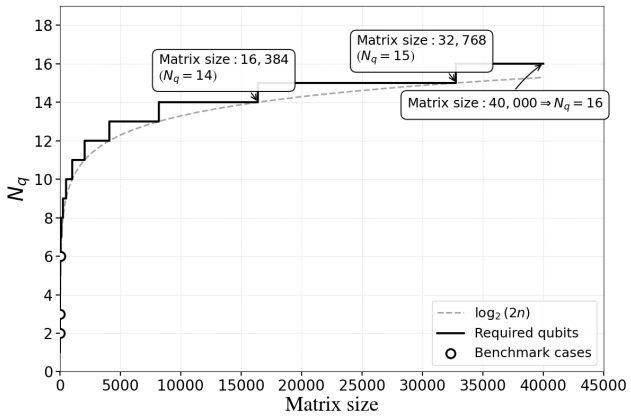


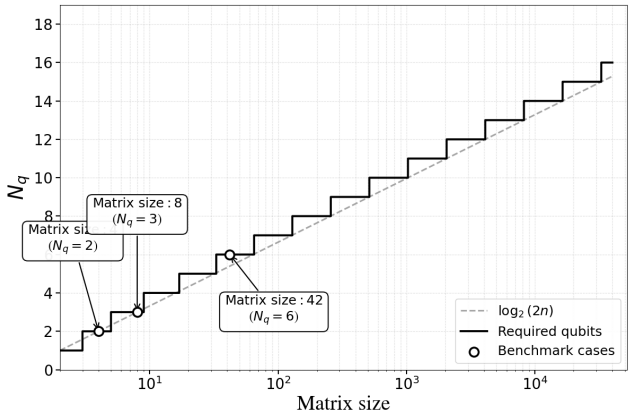
Figure 8.1: Comparison of the time and memory required by the classical approach to compute eigenvalues of a matrix of a given size, along with trend lines for each plot

8.2 Quantum computation scaling

Quantum computing requires a fundamentally different approach, as the matrices are stored within qubits. As illustrated in Fig. 8.2 the required qubit number scales logarithmically with the matrix size. For instance, a matrix of size $40\,000 \times 40\,000$ requires only 16 qubits. Therefore, with 100-qubit processors, one can theoretically encode matrices of dimensions up to $2^{100} \times 2^{100}$, far exceeding capabilities of any classical devices. In the 3D case of MST, where matrices are much bigger due to infinite ℓ that must be cut off at some ℓ_{max}, our approach still remains tractable and can accommodate systems with practically any number of scatterers. The computational time required to solve an eigenvalue of the matrix is mostly determined by the convergence of optimization loop. The number of VQD interactions is governed by the parameterization of the variational ansatz. Finally, the number of variational parameters scales with the number of qubits rather than count of scatterers. As a result, while moving to larger systems, adding more scatterers and increasing the angular momentum cutoff $\ell_{\max}$ does not significantly increase the computational cost. Nevertheless, the iterative procedure itself remains quite demanding, as demonstrated by the two-scatterer system (Fig. 6.6), which required more than 200 iterations for just one eigenvalue. Hybrid quantum-classical algorithms such as VQD offer a promising alternative to compute eigenenergies for systems that exceed the tractable limits of purely classical methods. More advanced hybrid algorithms, such as SQD, have already demonstrated the ability to model systems containing more than 12,000 atoms, which translates to effective matrix sizes of 10^{20} or more.[13,52] This benchmark required over 239 hours of quantum processor time, yielding more than 3 billion measurement results.



(a)



(b)

Figure 8.2: Scaling of the required number of qubits as a function of the secular matrix size $2n$, shown in linear (a) and (b) logarithmic scales.

9. Architecture and Tools of the QuantumMST Framework

All source code developed within the scope of this thesis, along with additional materials, is available on GitHub **QuantumMST**(`https://github.com/m-martyna/QuantumMST`) Its main purpose aligns with the focus of this study, which is to explore multiple scattering theory and its potential applications within the field of quantum computing. The entire code was developed and tested using Python 3.12.2 and Qiskit 2.4.1. The QuantumMST repository includes a utility library used to produce the results presented in chapters 6 and 7, along with usage examples that extend beyond them.

The eigenvalue computation scheme for a system at a given energy, illustrated in Fig. 9.1, divides the workflow into two stages. The first involves the positions of the scattering centers and the energy, which yields the structure constants, while the second one is performed for each individual scatterer to obtain all scattering matrices. Once these are determined, all components necessary to construct the secular matrix (Eq. (2.36)) are available, and its eigenvalues are computed.

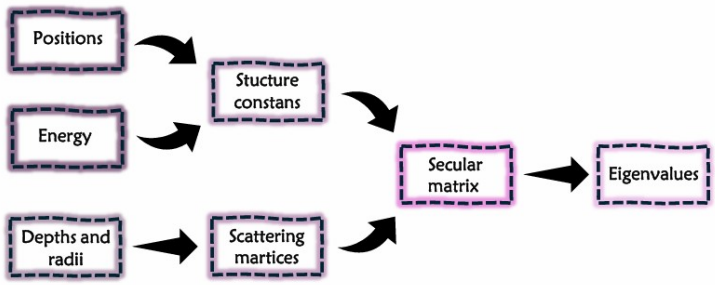


Figure 9.1: Workflow of the secular matrix eigenvalue calculation for a given system geometry and energy, illustrating the separation between structural and single-scatterer contributions

To determine the allowed energies, the procedure for constructing the secular matrix is repeated multiple times within the chosen solver. The solver requires the system geometry as input; in the case of NumPy and VQD based solvers the number of energy points used to interpolate the bound-state energies is also needed. As mentioned previously, the energy range extends from 0 to the depth of the deepest well. There is an added margin of 0.1 to avoid numerical instabilities. The approaches for

obtaining the bound-state energies are illustrated in the diagram Fig. 9.2.

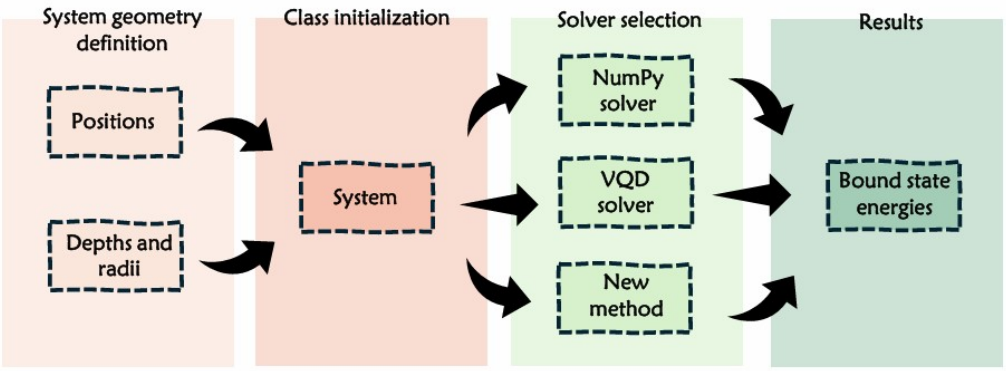


Figure 9.2: Visualization of the computational workflow for allowed energies in the QuantumMST library.

Aside from manual system definition, the library provides a set of pre-defined sample systems. In addition, it features system generators capable of creating system layouts given specified parameters. A practical example of these features is given in 9.1. The `scatterer` object receives the initial parameters as input and calculates all properties of a single scattering center. The class used to store all the system attributes is named `protein_structure` because this method may eventually be used to simulate proteins. However, this naming should not limit any interpretation or application of the method.

```
from QuantumMST import scatterer, protein_structure,
    generate_sysytem

#pre-difned sysytem
system_1 = protein_structure(reference_system = 1)

#manual definition
s1 = scatterer(r=0.8, V=-5)
s2 = scatterer(r=0.6, V=-6)
s3 = scatterer(r=0.7, V=-7)

system_2 = protein_structure(
    scatterers=[s1, s3, s2, s1],
    positions=[-1, 1, 3, 6]
)

#system generator
s, positions = generate_system(
    number=6,
    min_dist_pot=0.5,
    rmin=0.2,
    rmax=0.6,
    Vmin=-5.0,
    Vmax=-2.0,
    xmin=-5,
    xmax=5
)
system_3 = protein_structure(scatterers = s, positions = positions)
```

Listing 9.1: Three different possiblities of defining a system in QuantumMST

Regardless of its size, the potential profile of the selected system can be visualized (as shown in the plots in Fig. A). To do so, a method is first called to compute the potential array of the system with a chosen spatial resolution. Subsequently, calling the `plot_potential` function generates the final plot (as demonstrated in Listing 9.2).

```
from QuantumMST import plot_wavefunction, plot_potential

V, x_array = system_1.potential(resolution=100)
plot_potential(system_1)

system_1.energy = allowed_energies[0]
plot_wavefunction(x_array, V, system_1)
```

Listing 9.2: Ploting potential profile and wavefunction utilizing QuantumMST

The same listing highlights one of the most compelling features, the ability to verify the computed bound-state energies by plotting the corresponding wavefunctions. However, doing so requires assigning the calculated allowed energy to the system, which can be computed by `Solver` as shown in Listing 9.3.

```
from QuantumMST import Solvers

allowed_energies = Solvers(system_1, mode='numpy', nE=40)
```

Listing 9.3: Usage example of a solver for calculating bound-state energies in QuantumMST

More detailed usage (and always updated) examples can be found in the repository, along with the source code implementing the theoretical framework presented in section 2.2.3.

10. Future outlook

This project provides a solid foundation and presents considerable promise. However, there are numerous possible extensions worth implementing. Planned enhancements for the library include supporting arbitrary 1D potentials beyond square wells, as well as implementing self-consistent potential calculations to find the overall system energy. This would extend its capability from evaluating electronic properties under predetermined scattering potentials to directly calculating the potentials themselves.

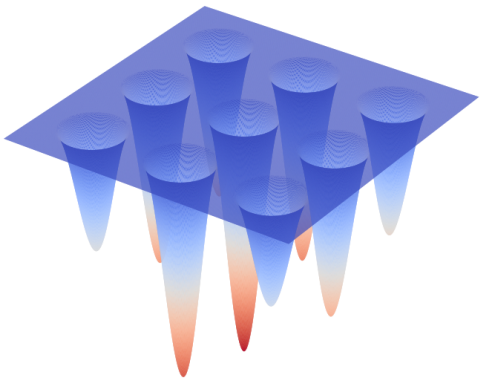


Figure 10.1: An example 2D representation of a system in the muffin-tin approximation

Only the one-dimensional MST formalism, which for n scatterers yielded $2n \times 2n$ matrices, is considered here. In this case, only ℓ equal to 0 or 1 is considered. However, the formalism to higher dimensions may be conceptually analogous, but significantly more complicated. The geometry of the system becomes more complicated, there are no longer simple wells, but one can truly see where the muffin-tin model took its name from (2D example Fig. 10.1). Consider, for instance the three-dimensional[19] case as an example, where infinite sums over the angular momentum index L appear, which denote orbital $\ell = 0, 1, ...$ and azimuthal $-\ell \leq m \leq \ell$ quantum numbers. ℓ must be cut off at a certain $\ell_{\max}$ to make the calculations feasible,

which yields the following number of basis $L = (\ell, m)$ elements:

$$\sum_{\ell=0}^{\ell_{\max}} (2\ell + 1) = (\ell_{\max} + 1)^2 \tag{10.1}$$

resulting in a total secular matrix dimension for n scattering centers in the form of:

$$n(\ell_{\max} + 1)^2 \times n(\ell_{\max} + 1)^2. \tag{10.2}$$

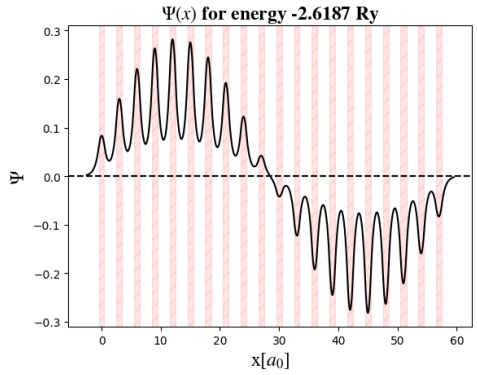
Importantly in MST, this dimension scales linearly with the size of the system, unlike traditional many-body approaches that scale exponentially.

Another potential extension involves fully periodic systems, corresponding to the Korringa-Kohn-Rostoker (KKR) formalism. As shown in 10.2 for a 20-scatterer periodic system, real-space wavefunctions can exhibit periodic features. However, standard KKR is obtained by applying a Fourier transform to the present framework. This transformation drastically reduces the matrices to the unit cell size, leaving no clear computational advantage for a quantum approach over classical methods. Consequently, such an approach could find potential application in systems with an exceptionally large number of atoms per unit cell. Regarding the 1D system, I intend to further develop a more refined variational algorithm that includes energy as a parameter. This involves testing an even wider variety of cost functions and analyzing their outcomes. As a result drastic reduction in computation time would enable computations for highly intricate systems.

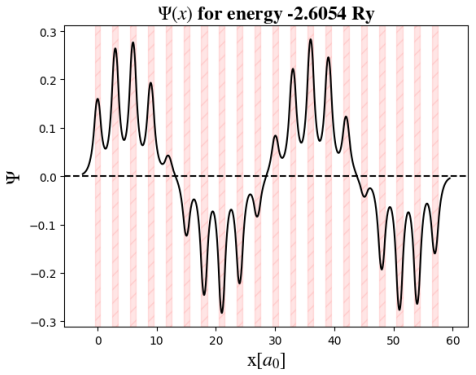
Furthermore, it is worth reconsidering the form of the secular matrix itself in order to better adapt it to the algorithms. The objective is to extract the maximum amount of information from a minimal size matrix. This would include exploiting symmetries and eliminating zero-valued entries while preserving the hermiticity of the matrix

It would also be valuable to compare the results with other methods, especially for increasingly larger systems. Even if these alternative approaches vary in their level of accuracy, such benchmarking would provide greater confidence in the results obtained.

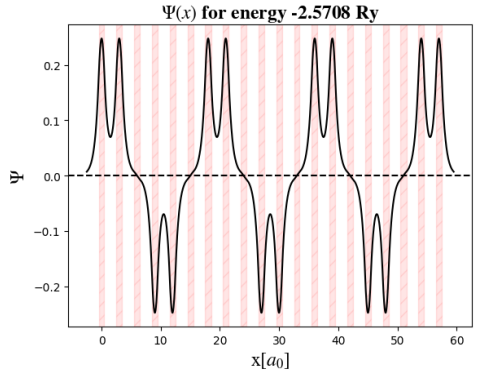
The ultimate goal, of course, would be to integrate this library into Qiskit, complete with usage examples, allowing anyone to utilize it as they see fit.



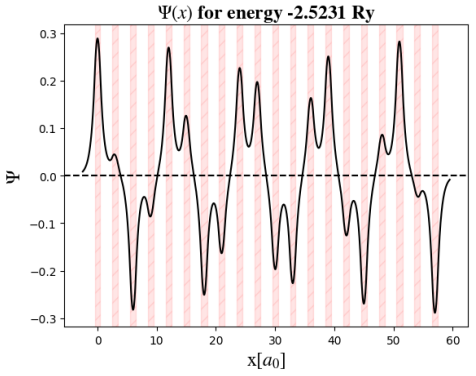
(a) first excited state



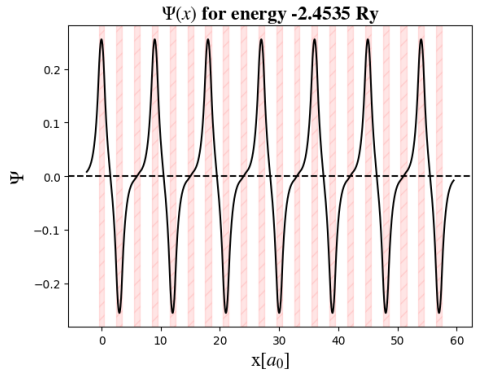
(b) third excited state



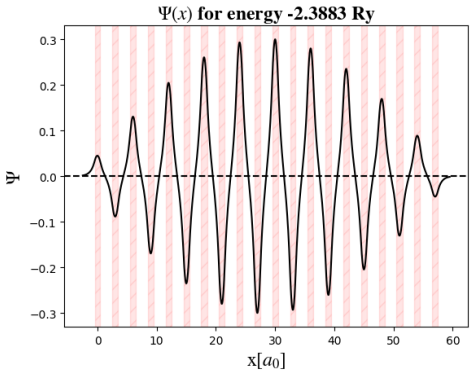
(c) 6th excited state



(d) 9th excited state



(e) 13th excited state



(f) 19th excited state

Figure 10.2: Wavefunctions for bound state energies computed for a periodic system of 20 potential wells with a well radius of $r = 0.5$, depth $V = -5.0$ and an inter-well separation of $d = 3.0$.

Summary

Quantum computing is a rapidly developing field in which many unknowns and unexplored territories remain. In the context of its application to solve problems using Multiple Scattering Theory, several key results were successfully achieved. In this work the one-dimensional formalism for MST with a definition of wavefunctions was developed and subsequently implemented as the library QuantumMST and made publicly available.

In addition to the conventional eigenvalue determination, the Variational Quantum Deflation algorithm was also successfully adapted to calculate the eigenvalues, which were then combined with the secular condition and interpolation to yield the bound state energies. Although accuracy decreased with increasing system size because a higher density of closely spaced eigenvalues caused errors in VQD performance, this algorithm has strong potential in terms of scalability. This is primarily due to the fact that the number of required qubits scales logarithmically with the matrix size. The impact of energy sampling on the approximation of allowed energies was investigated. An attempt was made to develop an algorithm that would drastically reduce the number of optimization iterations. Despite an initially promising concept, none of the tested variants of the cost function yielded consistent results across runs. Future development directions for the project were discussed in depth.

Despite some limitations, this study establishes that combining quantum computing and MST is a viable approach. It provides valuable insights into the practical limits of quantum algorithms and a solid foundation for future research. Unfortunately, until the full potential of quantum hardware can be realized, benefits over classical computers will remain out of reach. A basic algorithm such as VQD is insufficient and too computationally expensive. It is worth placing greater hope in new algorithms while treating VQD essentially as a baseline.

Bibliography

[1] IBM. *IBM Quantum Hardware*. International Business Machines Corporation. 2026. URL: https://www.ibm.com/quantum/hardware (visited on 08/12/2026).

[2] Richard P. Feynman. "Simulating Physics with Computers". In: *International Journal of Theoretical Physic* (1982). DOI: https://doi.org/10.1007/BF02650179.

[3] Oak Ridge Leadership Computing Facility. *Summit: OLCF's 200-Petaflop Supercomputer*. Oak Ridge National Laboratory. 2024. URL: https://www.olcf.ornl.gov/olcf-resources/compute-systems/summit/ (visited on 08/13/2026).

[4] Stephen Jordan. *Quantum Algorithm Zoo*. Comprehensive catalog of quantum algorithms. URL: https://quantumalgorithmzoo.org/ (visited on 08/13/2026).

[5] Lov K. Grover. *A fast quantum mechanical algorithm for database search*. 1996. arXiv: quant-ph/9605043 [quant-ph]. URL: https://arxiv.org/abs/quant-ph/9605043.

[6] P.W. Shor. "Algorithms for quantum computation: discrete logarithms and factoring". In: *Proceedings 35th Annual Symposium on Foundations of Computer Science*. 1994, pp. 124–134. DOI: 10.1109/SFCS.1994.365700.

[7] V. E. Elfving et al. *How will quantum computers provide an industrially relevant computational advantage in quantum chemistry?* 2020. arXiv: 2009.12472 [quant-ph]. URL: https://arxiv.org/abs/2009.12472.

[8] John Preskill. "Quantum Computing in the NISQ era and beyond". In: *Quantum* 2 (Aug. 2018), p. 79. ISSN: 2521-327X. DOI: 10.22331/q-2018-08-06-79. URL: https://doi.org/10.22331/q-2018-08-06-79.

[9] Seetharami Seelam et al. *Reference Architecture of a Quantum-Centric Supercomputer*. 2026. arXiv: 2603.10970 [cs.ET]. URL: https://arxiv.org/abs/2603.10970.

[10] IBM. *What is quantum-centric supercomputing?* IBM Think Topics. URL: https://www.ibm.com/think/topics/quantum-centric-supercomputing (visited on 08/18/2026).

[11] IBM. *What is high-performance computing (HPC)?* https://www.ibm.com/think/topics/hpc. Accessed: 2026-08-18.

[12] Youngseok Kim et al. “Evidence for the utility of quantum computing before fault tolerance”. In: *Nature* 618.7965 (2023), pp. 500–505. ISSN: 1476-4687. DOI: 10.1038/s41586-023-06096-3. URL: https://doi.org/10.1038/s41586-023-06096-3.

[13] Jr. Kenneth M. Merz et al. *Crossing the 12,000-atom barrier with heterogeneous quantum-classical supercomputing: quantum chemistry of protein-ligand complexes*. 2026. arXiv: 2605.01138 [quant-ph]. URL: https://arxiv.org/abs/2605.01138.

[14] Simon Martiel et al. *Sampling hard circuits with verifiably high fidelity*. 2026. arXiv: 2607.25941 [quant-ph]. URL: https://arxiv.org/abs/2607.25941.

[15] Eyal Leviatan et al. *Resolving Structure in Prethermal Floquet Dynamics with Precision Quantum Computation*. 2026. arXiv: 2607.24937 [quant-ph]. URL: https://arxiv.org/abs/2607.24937.

[16] Samantha V. Barron et al. *Observable Estimation in the Absence of Classical Verification*. 2026. arXiv: 2607.25998 [quant-ph]. URL: https://arxiv.org/abs/2607.25998.

[17] IBM Quantum. *Quantum Advantage Tracker*. IBM. URL: https://quantum.ibm.com/advantage-tracker (visited on 09/07/2026).

[18] Lord Rayleigh. In: *Philosophical Magazine* 34 (1892). *The Theory of Sound*, 2nd Ed. vol. 2, p. 328 (1892) (reprinted by Dover Publications, New York (1976)), p. 481.

[19] Antonios Gonis and William H. Butler. *Multiple Scattering in Solids*. 1st ed. Graduate Texts in Contemporary Physics. New York, NY: Springer New York, NY, 2000. ISBN: 978-0-387-98853-5. DOI: 10.1007/978-1-4612-1290-4.

[20] W. H. Butler. “One-dimensional model for transition metals and their alloys”. In: *Phys. Rev. B* 14 (2 July 1976), pp. 468–478. DOI: 10.1103/PhysRevB.14.468. URL: https://link.aps.org/doi/10.1103/PhysRevB.14.468.

[21] P. W. Anderson. “More Is Different”. In: *Science* 177.4047 (1972), pp. 393–396. DOI: 10.1126/science.177.4047.393. eprint: https://www.science.org/doi/pdf/10.1126/science.177.4047.393. URL: https://www.science.org/doi/abs/10.1126/science.177.4047.393.

[22] Tomasz Maziarka. “Idea emergencji – zarys ogólny”. pl. In: *Zagadnienia Filozoficzne w Nauce* LII (2013), pp. 131–177.

[23] Ben Brubaker. *Complexity Theory’s 50-Year Journey to the Limits of Knowledge*. https://www.quantamagazine.org/complexity-theorys-50-year-journey-to-the-limits-of-knowledge-20230817/. Accessed: 2026-06-30. Aug. 2023.

[24] Nikhil Srivastava. “The Complexity of Diagonalization”. In: *Proceedings of the 2023 International Symposium on Symbolic and Algebraic Computation*. ISSAC ’23. Tromsø, Norway: Association for Computing Machinery, 2023, pp. 1–6. ISBN: 9798400700392. DOI: 10.1145/3597066.3597145. URL: https://doi.org/10.1145/3597066.3597145.

[25] Ashley Montanaro. *Computational Complexity: Lecture Notes (Part III)*. https://people.maths.bris.ac.uk/~csxam/teaching/cc-lecturenotes.pdf. Michaelmas term lecture notes, DAMTP Cambridge. 2012.

[26] John Watrous. *Quantum Computational Complexity*. 2008. arXiv: 0804.3401 [quant-ph]. URL: https://arxiv.org/abs/0804.3401.

[27] Melanie Mitchell. *Complexity: A Guided Tour*. Oxford: Oxford University Press, 2009. ISBN: 978-0195124415.

[28] IBM Quantum Learning. *Quantum Circuits*. IBM. 2024. URL: https://quantum.cloud.ibm.com/learning/en/courses/basics-of-quantum-information/quantum-circuits/circuits (visited on 07/01/2026).

[29] Nobel Prize Outreach. *The Nobel Prize in Physics 2022 - Popular science background: How entanglement has become a powerful tool*. 2022. URL: https://www.nobelprize.org/prizes/physics/2022/popular-information/ (visited on 07/01/2026).

[30] IBM Quantum. *Measure qubits*. https://quantum.cloud.ibm.com/docs/en/guides/measure-qubits. Accessed: 2026-07-06.

[31] IBM Quantum. *Quantum Diagonalization Algorithms*. https://quantum.cloud.ibm.com/learning/en/courses/quantum-diagonalization-algorithms. IBM Quantum Learning Course; accessed: 2026-06-29. 2026.

[32] Changwon Lee and Daniel K. Park. *Generative Krylov Subspace Representations for Scalable Quantum Eigensolvers*. 2025. arXiv: 2512.19420 [quant-ph]. URL: https://arxiv.org/abs/2512.19420.

[33] Cristian L. Cortes and Stephen K. Gray. "Quantum Krylov subspace algorithms for ground- and excited-state energy estimation". In: *Physical Review A* 105.2 (Feb. 2022). ISSN: 2469-9934. DOI: 10.1103/physreva.105.022417. URL: http://dx.doi.org/10.1103/PhysRevA.105.022417.

[34] Ana Martin Delgado et al. "Sample-Based Krylov Quantum Diagonalization". In: *PRX Quantum* 4 (4 Nov. 2023), p. 040331. DOI: 10.1103/PRXQuantum.4.040331. URL: https://link.aps.org/doi/10.1103/PRXQuantum.4.040331.

[35] Colin D. Bruzewicz et al. "Trapped-ion quantum computing: Progress and challenges". In: *Applied Physics Reviews* 6.2 (May 2019). ISSN: 1931-9401. DOI: 10.1063/1.5088164. URL: http://dx.doi.org/10.1063/1.5088164.

[36] Guido Burkard et al. "Semiconductor spin qubits". In: *Reviews of Modern Physics* 95.2 (2023). ISSN: 1539-0756. DOI: 10.1103/revmodphys.95.025003. URL: http://dx.doi.org/10.1103/RevModPhys.95.025003.

[37] Dawei Lu et al. *NMR quantum information processing*. 2015. arXiv: 1501.01353 [quant-ph]. URL: https://arxiv.org/abs/1501.01353.

[38] Chetan Nayak et al. "Non-Abelian anyons and topological quantum computation". In: *Reviews of Modern Physics* 80.3 (2008), pp. 1083–1159. ISSN: 1539-0756. DOI: 10.1103/revmodphys.80.1083. URL: http://dx.doi.org/10.1103/RevModPhys.80.1083.

[39] Tzu-Ching Yen, Vladyslav Verteletskyi, and Artur F. Izmaylov. *Measuring all compatible operators in one series of a single-qubit measurements using unitary transformations*. 2020. arXiv: 1907.09386 [quant-ph]. URL: https://arxiv.org/abs/1907.09386.

[40] Jay Gambetta. *What are Qiskit Primitives?* Qiskit / Medium. May 2022. URL: https://medium.com/qiskit/what-are-qiskit-primitives-9bf63c1eacc7 (visited on 09/20/2026).

[41] IBM Quantum. *Get Started with Sampler*. IBM Cloud Documentation. URL: https://quantum.cloud.ibm.com/docs/en/guides/get-started-with-sampler (visited on 09/21/2026).

[42] Qiskit Community. *Qiskit Algorithms Optimizers Documentation*. https://qiskit-community.github.io/qiskit-algorithms/apidocs/qiskit_algorithms.optimizers.html. Accessed: 2026-07-07. 2026.

[43] SciPy Community. *scipy.optimize.minimize — SciPy Reference Guide*. https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.minimize.html. Accessed: 2026-07-07. 2026.

[44] IBM Quantum. *PauliEvolutionGate*. IBM. 2026. URL: https://quantum.cloud.ibm.com/docs/en/api/qiskit/qiskit.circuit.library.PauliEvolutionGate (visited on 09/20/2026).

[45] Timothy N Georges et al. "Pauli decomposition via the fast Walsh-Hadamard transform". In: *New Journal of Physics* 27.3 (Feb. 2025), p. 033004. DOI: 10.1088/1367-2630/adb44d. URL: https://doi.org/10.1088/1367-2630/adb44d.

[46] IBM Quantum. *Simulate quantum circuits with Qiskit Aer*. Accessed: 2026-09-21. IBM. 2024. URL: https://quantum.cloud.ibm.com/docs/en/guides/simulate-with-qiskit-aer.

[47] Qiskit Contributors. *EfficientSU2: Qiskit API Documentation*. IBM Quantum. URL: https://quantum.cloud.ibm.com/docs/en/api/qiskit/qiskit.circuit.library.EfficientSU2 (visited on 09/15/2026).

[48] W. H. Butler and R. K. Nesbet. "Validity, accuracy, and efficiency of multiple-scattering theory for space-filling scatterers". In: *Phys. Rev. B* 42 (3 July 1990), pp. 1518–1525. DOI: 10.1103/PhysRevB.42.1518. URL: https://link.aps.org/doi/10.1103/PhysRevB.42.1518.

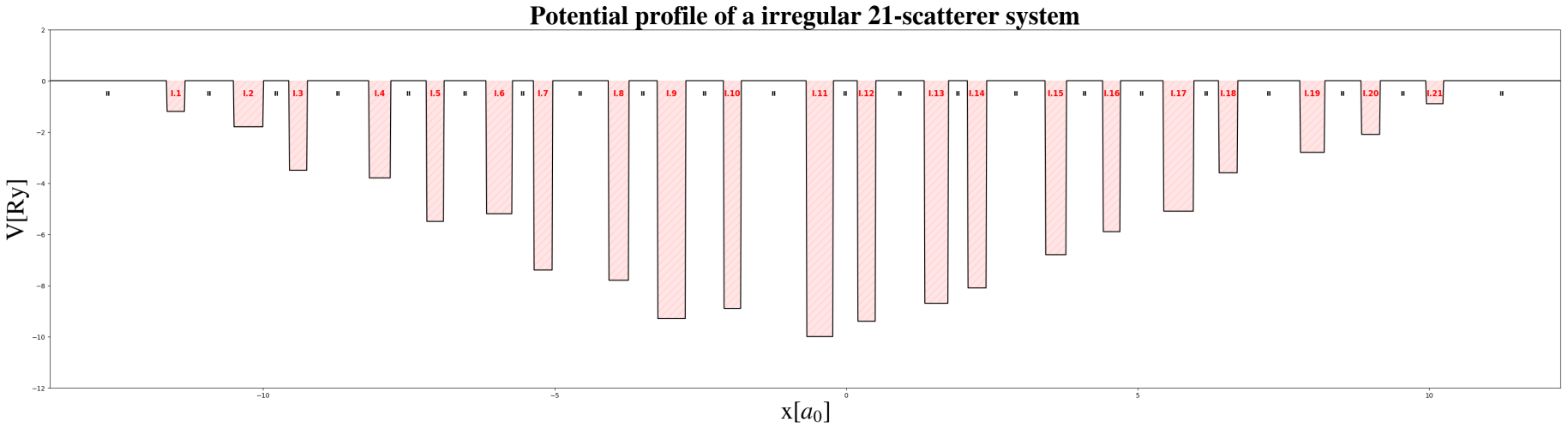
[49] Qiskit Development Team. *Qiskit 1.0 Documentation: Quantum Information (qiskit.quantum_info)*. https://quantum.cloud.ibm.com/docs/en/api/qiskit/1.0/quantum_info. Accessed: 2026-09-21. 2024.

[50] *numpy.linalg.eigvals — NumPy v1.26 Manual*. https://numpy.org/doc/stable/reference/generated/numpy.linalg.eigvals.html. [Accessed: June 28, 2026]. 2026.

[51] *LAPACK — Linear Algebra Package*. https://www.netlib.org/lapack/. [Accessed: June 28, 2026]. 2026.

[52] IBM Quantum. *Quantum-centric supercomputing simulates 12,635-atom protein*. May 2026. URL: https://www.ibm.com/quantum/blog/cleveland-clinic-riken-chemistry (visited on 09/14/2026).

A. Potential Distribution of the 21-Scatterer System



List of Figures

4.1 The Bloch sphere, providing a geometric representation of a single-qubit state. 22
4.2 Quantum circuit for the generation of a Bell state $|\Phi^+\rangle$, containing 2 qubits and 2 classical bits for measuring the results. 24
4.3 Diagram of the VQE alogithm, highlighting the input/oputput data and the division of tasks between the CPU and the QPU. 26
4.4 Qubit lattice of the ibm_kingston backend 27

5.1 Imaginary parts of eigenvalues for a system of (a) 4 and (b) 21 scatterers. 30

6.1 One-dimensional potential profiles for configurations with (a) two and (b) four scatterers. The geometry of the two-scatterer setup is adopted from Butler[20] to validate the results obtained in this study against benchmark literature data. Dashed red regions indicate areas inside the potential, with each being assigned a label ($I.1$, $I.2$, etc.), whereas the interstitial regions are all designated as II. Following the form of Eq. (2.3), Rydberg atomic units are used, where the potential is given in Ry and the position in a_0. 33
6.2 Close-up views of the energy-dependent secular determinant for systems with different numbers of scatterers: (a) two scatterers, (b) four scatterers, (c) 21 scatterers, and (d) again for 21 scatterers, illustrating the numerical explosion. 34
6.3 Eigenvalue spectra as a function of energy for varying system sizes: (a) two scatterers, (b) four scatterers, (c) 21 scatterers and (d) corresponding close-up view. Additionally, for systems with more than two scatterers, the plots feature a color gradient distinguishing individual eigenvalue, along with the total number of eigenvalues displayed on the side. 35
6.4 Plot showing only the eigenvalues selected for polynomial interpolation for the 21-scatterer system. 36
6.5 Schematic representation of the parameterized `EfficientSU2` ansatz. 37
6.6 Optimization process across a single VQD run for the first four eigenvalues using different Qiskit optimizers for the system with two scatterers. 38
6.7 Comparison of eigenvalues computed using VQD and NumPy for the two-scatterer system at a chosen energy resolution. Alongside the eigenvalue comparison, the absolute error relative to the NumPy calculations is displayed. 39

6.8 Relative error of the interpolated bound states obtained by VQD and NumPy (with respect to high-accuracy reference values computed by NumPy for $1,000$ energy points) as a function of the number of energy points used in calculation for the two-scatterer system, showing both the full-scale plots and magnified views. 40
6.9 Relative error of the interpolated bound states obtained via VQD and NumPy, computed relative to high-accuracy reference values computed by NumPy for $1,000$ energy points, versus the number of energy points used in calculation for the four-scatterer system. 40
6.10 Comparison between VQD and NumPy eigenvalues for the four-scatterer system at a given energy resolution. The absolute error relative to the NumPy reference values is also shown. 41
6.11 Wavefunctions and their squared moduli for bound-state energies of the benchmark two-scatterer system. 42
6.12 Wavefunctions and their squared moduli for selected bound-state energies of the benchmark four-scatterer system. 43
6.13 Selected wavefunctions and their squared moduli for bound-state energies of the benchmark 21-scatterer system. 44

7.1 Optimization trajectories of the proposed algorithm attempting to find the bound state energy for parameters $t = 0.1$ and $\beta = 0.5$, where (a) convergence to the first excited state, (b) descent toward the ground state that stops prematurely, (c) a failed run converging to an irrelevant value, and (d) convergence to the ground state. . . 47
7.2 Example heatmaps showing the accuracy of the custom algorithm's results when run with various β and t parameters. 47

8.1 Comparison of the time and memory required by the classical approach to compute eigenvalues of a matrix of a given size, along with trend lines for each plot 49
8.2 Scaling of the required number of qubits as a function of the secular matrix size $2n$, shown in linear (a) and (b) logarithmic scales. 50

9.1 Workflow of the secular matrix eigenvalue calculation for a given system geometry and energy, illustrating the separation between structural and single-scatterer contributions 51
9.2 Visualization of the computational workflow for allowed energies in the QuantumMST library. 52

10.1 An example 2D representation of a system in the muffin-tin approximation 54
10.2 Wavefunctions for bound state energies computed for a periodic system of 20 potential wells with a well radius of $r = 0.5$, depth $V = -5.0$ and an inter-well separation of $d = 3.0$. 56

List of Tables

4.1 Examples of quantum gates with their symbols and properties 23

6.1 Summary of parameters including scatterer radii (r_i), well depths (V_i) and center positions (x_i), for each of the benchmark systems. 34
6.2 Interpolated allowed energies, along with the adjacent energy points bracketing each sign change identified for the 21-scatterer configuration. 36
6.3 Quantum resource requirements: secular matrix dimensions, qubit counts and Pauli term scaling across benchmark scatterer configurations. 37
6.4 Comparison of calculated energy levels across Butler's calucations,[48] classical and quantum methods for the two scatterer benchmark system. 38